

# **ASIS: Accenture Voice** **Assistant Final Design** **Document**

Group L07

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# **Executive Summary**

The ASIS Accenture Lab Assistant stands as the cornerstone of streamlined efficiency within the research landscape, offering an integrated platform where all areas of laboratory operations converge seamlessly. At the heart of this project's goal lies the resolve to address the longstanding challenge identified by Accenture: the complexities caused by disparate, proprietary technologies that govern the operation of expensive laboratory equipment from various manufacturers. This often hinders interoperability and impedes collaborative efforts.

As the computer science team on this project, we are tasked with orchestrating a transformative solution and have embarked on a path to harness the potential of the cutting-edge technologies Accenture has to provide for us. Leveraging the intuitive UI environment of Lab Voice, alongside tools like Scitara and LabVantage LIMS, the team set out to engineer a platform that not only bridges the gap between disparate technologies, but additionally empowers researchers with accessibility and control.

A key component to this endeavor is the seamless integration of voice commands, affording researchers the ability to initiate laboratory processes easily from the moment they step foot into the lab. Through these voice interactions, researchers can swiftly navigate complex workflows, trigger equipment functionality, and access vital information, all with the natural power of their voice.

Not only does the ASIS Accenture Lab Assistant represent more than just connectivity, it is a game changer in the field of research by making lab operations easy for everyone. By simplifying complex systems and creating a unified environment, it gives researchers of all levels a user-friendly interface that encourages collaboration, speeds up experiments, and sparks innovation. As Accenture's commitment to innovation, the Lab Assistant leads the charge in using technology to transform research, pushing boundaries and opening new paths to discovery.

In our quest for development, we have gone through extensive research and traversed a multitude of avenues in order to gain insights and knowledge essential for crafting the ASIS Accenture Lab Assistant. Our journey began with the team engaging in informal discussions with the sponsor and esteemed members of the Accenture team. These sessions served as valuable opportunities to get firsthand information, uncovering nuances and intricacies that would shape the trajectory of our team development's.

However, our pursuit for knowledge didn't stop there, we delved deeper by immersing ourselves in formal training sessions dedicated to mastering the intricacies of Lab Voice. Through these structured learning experiences. We were able to build our skills with this platform and lay the groundwork for seamless integration with this framework in the future. After recognizing the interconnectedness of technologies in the lab ecosystem, we reached out to Scitara and data processing experts, engaging in insightful dialogues and knowledge-sharing sessions. This gave us rich insight into valuable context that gave us a holistic understanding for effective development.

Furthermore, we gained important hands-on experience after venturing into the heart of the laboratory itself. Through multiple visits, we familiarized ourselves with the myriad of instruments in the Accenture lab, immersing ourselves in their complexities and unraveling their functionalities. In essence, our research journey and pursuit of knowledge and understanding provided the team with the knowledge required for starting this wonderful project.

# **Introduction**

## **Project Identification/Overview**

The project is to develop a voice assistant integrated into a “showcase” laboratory owned by Accenture and located in Orlando. The specifics of this laboratory are elaborated upon later, in the “Acknowledgments -> Accenture Orlando Laboratory” section of this document.

More specifically, this voice assistant is to be integrated with the laboratory such that it can perform a set sequence of tasks, at a minimum. These tasks involve various laboratory systems and subsystems, such as instruments, the laboratory’s mixed LIMS<sup>1</sup> and ELN<sup>2</sup>, and scientists’ personal devices, especially mobile devices (phones or tablets). Some of the instruments in the lab are accompanied by proprietary full-stack locally hosted software applications that run on a Windows computer connected (usually – and as is the case in this particular laboratory – by a wired connection) directly to the instrument in question. Others have only the software used to make the instrument emit basic readings. These require more complicated means – such as IoT<sup>3</sup> gateway boxes – to be communicated with or commanded by remote software, in other words without physical interaction with the devices’ hardware.

The sponsor would like to see that the voice assistant can be coordinated with their Orlando laboratory’s existing systems and subsystems such that a set sequence of events can be performed. These systems,

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<sup>1</sup> Laboratory Information Management System

<sup>2</sup> Electronic Laboratory Notebook

<sup>3</sup> Internet of Things

subsystems, and the minimum desired set sequence of events will be explained in further detail in later sections of this document. To accomplish this plethora of objectives, a variety of tools, technologies, and products are to be employed. These tools, technologies, and products will be explained in greater depth later.

## What is a LIMS?

A LIMS, or Laboratory Information Management System, is used by modern scientific laboratories – especially in the heavily regulated pharmaceutical industry – to keep track of and maintain detailed records of data from and activities performed in a company's or other organization's laboratories. These are vital to many industry clients of the sponsor's, because many are engaged in heavily regulated industries such as pharmaceuticals. In the industries in which the sponsor works and in which its clients work, keeping very detailed records in all respects is considered vital to ensuring regulatory compliance and maintaining consumer confidence. As a result, many companies rely on records called "audit trails," which are step-by-step records of data collections, measurements, experiments, and miscellaneous other processes performed in the laboratory. For many organizations, creating and maintaining such records is – while of vital importance – extremely time-consuming and tedious without automated support, as each step in a sequence must be extensively documented and stably preserved.

Consequently, many LIMS providers include a feature to automatically create and store these audit trails. This is part of why the sponsor finds it of paramount importance to integrate the voice assistant – albeit it is not necessary to do so for the MVP – with a LIMS, and more specifically, to do so

with the LIMS used to manage the information from their Orlando laboratory. This will be elaborated upon further later in this document.

LIMSs differ from ELNs, which will be explained in the next subsection. Specifically, in that LIMSs are more for sample tracking (for example, tracking a particular batch of 96 samples of dichloroethenes, or perhaps – what the sponsor typically works with – some kind of food coloring), metadata tracking (for instance, when a batch of liquid samples had the absorbances of its samples measured by a plate reader – a workflow that the sponsor has demonstrated several times to the team), and process and workflow documentation. ELNs, on the other hand, are more often for recording experiment data and other immediately relevant data, rather than metadata necessary for regulatory compliance. Both can be used for regulatory compliance, but a LIMS is more specifically designed for that purpose and exquisite record-keeping, whereas an ELN is more intended to facilitate scientific research processes and efforts.

## What is an ELN?

An ELN, or Electronic Laboratory Notebook, is a way to store results and data such that it can be electronically accessed. It usually provides a centralized database for electronic information storage for members of the laboratory's organization. Most modern ELNs default to storing this data in the cloud, and a very common cloud choice among modern ELN providers is in the AWS cloud (as is the case with the ELN that the sponsor possesses). ELNs can be used to store both structured – meaning consistently formatted, such as the data automatically output by a machine when a user commands a diagnostics test to be run – or unstructured data, such as manually entered experiment data or notes.

Usually, free text is employed, and the formatting of such text is guided by the ELN itself, the organization that uses the ELN, or simply by the users. This is a prime example of unstructured data. However, oftentimes, a laboratory's ELN is integrated with – or at least receives data, in the form of files, from – instruments in the laboratory, assuming that the instruments provide such capabilities. Frequently, modern laboratory instruments come with proprietary full-stack (meaning they have a local database, UI, and back end, although not always – and in fact often not – an accessible API for interaction with by developers and other programs), locally hosted Windows applications that run on a Windows machine connected directly (usually by a wired connection) to the instruments.

Moreover, these instruments and their software are usually outdated and – quite often – incompatible with modern interconnectivity and data sharing technology, like cloud services such as Google Sheets or Box. Nevertheless, these instruments generate vast quantities of data, and these data or measurements are of great value to the scientists that use these instruments, as is storing them, so the instruments need some way to reliably output storable, consistently formatted data. Consequently, these instruments tend to output files, most often .csv files, .txt files, or .xls, .xlt, .xlm, or .xlsx (Excel) files, and rely on a more antiquated method of sharing said files to their connected Windows machines – Windows File Share<sup>4</sup>. These files are frequently then retrieved and stored directly as attachments into a laboratory's ELN. Automating the process of data storage from instrument-connected computers is a large component of the sponsor's overarching mission, as well as the missions of their clients (hence the desire to integrate the sponsor's bundled LIMS and ELN service with at least

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<sup>4</sup> WFS



on of their instruments' workflows and the voice assistant that can be used to trigger such a workflow remotely).

## Bundled LIMS/ELN Services

ELNs are often used to store more than merely experimental data nowadays. In fact, oftentimes in more recent years, ELNs have begun to provide much of the same functionality as LIMSs, such as providing automated audit trail collection. Therefore, many laboratory electronic system providers have decided to bundle their LIMS and ELN services into a single service. Such is the case for the bundled ELN/LIMS service utilized by the sponsor's Orlando laboratory. Their choice ELN/LIMS provider is LabVantage, which is expounded in greater detail in the "LabVantage LIMS/ELN" subsection.

## Project Significance

The sponsor of this project is a team of pharmacological chemistry scientists, technical developers, technical analysts, and miscellaneous other team members at Accenture, a multinational business consulting company, researching ways to streamline the modern laboratory workflow through various means. Chief among these means, for the purposes of this project, is the goal of automation and automating processes in a typical modern pharmaceutical laboratory, worded as "hands-free science" by the Accenture team program lead. Automation is the means to achieve one of the sponsor's overarching goals, which is efficiency. Firstly, it will help to reduce the amount of time that scientists working in such laboratories spend performing

low-value work (such as manually pipetting liquid samples into a 96-well plate, one well at a time, or setting up and preparing and then finally running instruments to perform necessary chemical calculations), so that these scientists – who are typically highly trained, very intelligent, and very capable individuals – can spend more time performing high-value, critical, knowledge-based thinking work instead of menial tasks that can be perhaps more easily, and even more quickly, performed by a machine.

Another of the sponsor's goals that they hope to achieve through automation is to reduce – and perhaps, ultimately, completely eliminate – the need for scientists to be physically present in their laboratories and with their instruments. This will enable both greater interregional and international collaboration (someone in France can check on the status of or control a lab in Oak Ridge, Tennessee, for example), as well as potentially reduce costs, because Labs may not need to be designed for human occupation and comfort.

Yet another major goal of the sponsor's, one which is not necessarily solved by automation, is more nuanced and is a product of the current state of the modern chemical lab instruments software space. Take the example of Apple, which creates its own ecosystem for all of its products whereby cross-compatibility with other operating systems, file systems, and whatnot goes unsupported – or, at best, poorly supported. Similarly, many of the largest laboratory instrument manufacturers today – including Hamilton, Thermo-Fisher, Mettler-Toledo, and BioTek, to name a few (and let it be noted that the sponsor possesses at least one instrument from each of these manufacturers in their laboratory) – have designed and created their respective software suites and products such that they are largely inaccessible to other instruments and elements outside of their company's ecosystem. So, for example, if a group of scientists wants to use a

Thermo-Fisher thermocycler (a type of chemistry instrument that the Accenture team owns) in their laboratory, and wants to automate said laboratory using the software built into that Thermo-Fisher instrument, then one would need to convert nearly every instrument in their entire laboratory into a Thermo-Fisher instrument. This, as the sponsor has expounded, is not only tremendously inconvenient, but also is very often simply impractical. Very frequently, chemistry laboratories of all kinds – not only in the pharmaceutical industry – must incorporate instruments from a variety of different manufacturers into their laboratories, experiments, and workflows. This could occur for a number of reasons, from cost prohibitions to, simply, one manufacturer making a superior version of the particular instrument desired versus the version made by another manufacturer.

This illustration of the restrictive software ecosystems put in place by modern chemical laboratory instrument manufacturers is to elucidate another one of the sponsor's major goals. Ultimately, the sponsor desires to essentially unify and centralize the disparate software packages and suites in modern chemistry laboratory instruments into a single system that can be easily accessed and controlled from anywhere. They seek to design and provide a proof of concept of a fully automated, fully integrated, and successfully modularized – from a software perspective – modern pharmacological chemistry laboratory, in order to reduce costs and increase laboratory efficiency. The LabVoice voice assistant is crucial to helping to accomplish that – namely, to achieving the goal of “hands-free science”. (If one can command it with their voice, with no hands-on interaction, then it is, by definition, “hands-free.”)

## **Personal Motivations**

Edrienne Gregana

I've always had this curiosity for the field of artificial intelligence (AI) and how much potential it had, which is what drew me to the ASIS: Voice Assistant. The idea of delving into the intricate web of natural language processing, machine learning algorithms, and human-computer interaction intrigued me. With how prevalent and increasingly relevant artificial intelligence is becoming in modern technology, I knew that this senior design project would be my opportunity to immerse myself in this dynamic field and contribute to the advancement of such a useful piece of technology.

While the majority of my software development background lies in web development, particularly front-end development, I saw this project as a great way to expand my horizons and venture into unfamiliar territory. The appeal of exploring the capabilities of voice assistant technology, beyond the usual products like Alexa, Siri, and Google Home, has inspired me to be part of this project which is both innovative and impactful. Beyond the technical intricacies, I am excited about the prospect of collaborating with experts from different disciplines, including data science. By embracing this interdisciplinary approach, I get to learn more about the intersectionality of technology with other fields and how wide of a scope technology covers. Not only is it wonderful to be able to team up with students from other fields, but it's also exciting to be able to work with Accenture and the advanced

technologies that they hold in their labs. I am eager to challenge myself, step out of my comfort zone, and cultivate a deeper understanding of artificial intelligence.

Before committing to a Computer Science major, my academic journey initially led me towards a pursuit in health sciences, fueled by a lifelong aspiration to become a medical professional. The desire to positively impact the lives of others has always been a driving force in my career aspirations. However, upon delving deeper into the field of Computer Science, I found myself increasingly drawn to the intersection of technology and healthcare. This shift in focus developed from a realization of the potential for innovation and advancement within health-related industries through the application of technology. The prospect of leveraging computational tools and techniques to address healthcare challenges and improve patient outcomes resonated deeply with my passion for both science and technology. The ASIS: Voice Assistant project is the perfect intersection of my interests, seamlessly blending the realms of applied sciences and technology. By developing a solution that enhances efficiency and productivity in laboratory environments, I have the opportunity to contribute to advancements in scientific research.

Expanding on my previous point, my initial experience with the realm of artificial intelligence was marked by a lack of full understanding of its scope and capabilities. While I knew about artificial intelligence in a basic sense and had seen it in action through things like ChatGPT, I didn't quite grasp just how common and impactful it had become. This is what fueled my interest in diving deeper into this project and is what contributed to me choosing it as my number one pick. I was amazed to discover how artificial intelligence is practically everywhere nowadays, from predictive analytics in businesses to automation on factory floors. Seeing its wide-ranging

applications made me realize the importance of actually learning about artificial intelligence. So, armed with this newfound curiosity, I became committed to exploring this fascinating field and uncovering its potential through the Accenture ASIS: Voice Assist project.

Ultimately, I aspire to leverage this experience to not only help me advance in my knowledge and career but also as a way to contribute to something meaningful and hopefully useful in the trajectory of the future. I chose this project because I am confident that this is something that would be beneficial to have, especially in the lab where cross-contamination is so prevalent. Mitigating the complexities inherent in manual tasks and promoting seamless workflow automation, would alleviate the burdens faced by researchers and also catalyze scientific discoveries and breakthroughs. In essence, I am driven by the belief that this project will have the potential to be transformative.

## Initial Ideas

Beyond our sponsors' requirements, there are a few little features I was considering. For instance, I wanted to add a way to make sure that only authorized personnel can use the voice assistant. We're also thinking about adding progress tracking so researchers can just use their voice to monitor the experiment's progress without having to walk over to the other side of the lab and checking. In addition to the progress checks, maybe there could be an implementation of an inventory manager that lets the team know when supplies are running low. This could help in scenarios where someone forgets to check the materials and allows the team to be prepared ahead of time. Also, a safety alert/reminder would be useful to add whenever a process with a dangerous machine or substance is initiated. This idea is more of a further stretch but collaborative tools through voice-activated messaging could also be useful in the future. This would allow researchers to

file share with their fellow teammates through a simple command. While our project already has its set of challenges, such as linking the software to their lab machines and whatever database they use to do calculations, I think having those little details would be useful if they're not already implemented. The ultimate goal of all these features is to just make the lab work easier and more efficient for researchers.

## Christopher Fong

As the landscape for jobs rapidly changes, implementation of artificial intelligence has become increasingly demanded and adopted by employers. I didn't want to fall behind in the job market in the future when I graduate because of my lack of exposure with utilizing AI in past projects. I took my first steps into learning about ways it can be utilized by creating my own voice assistant with some similarities to the VoiceAssist program that we will be developing in our project for Accenture. I thought my experience in making something similar would be helpful to the team, and I'd like to be able to see how real business settings would take my knowledge and integrate it into their workflow. With that in mind, I chose this project as my first choice.

What I was hoping to mainly work on with Accenture on the ASIS VoiceAssist project when I selected it is the implementation of data processing utilizing large language models and speech-to-text. Figuring out how to connect pipelines from point A to point B to transform information in a way that generates value has been something I've always enjoyed because it's like designing what I call a "well-oiled machine" most of the time. This doesn't always end up being the case especially when resource constraints exist, but trying to reach the goal of having that is what makes it enjoyable

to me. The most fascinating part of this project for me is likely going to be bringing it all together because I want to learn what steps I can take to design something for my own home automation for example. Many proprietary solutions already exist on the market, but being able to pull from open-source projects and improving them with my own capabilities would be far more useful for myself and breaking into the market for a potential position working with a company that specializes in home automation. There are also benefits of working with Accenture because of the connections that are created by working with such a large company while already having much of the knowledge I need to successfully deliver this project by the end of Summer semester.

Overall, I've always had a large interest in how I can make life easier for people by using AI and automation since I discovered the concepts. Bringing these ideas to reality is what drives me to complete this project as flawlessly as I can, but my one hope I have is that LabVoice isn't too much of a black box which will lower my exposure to the complexities I expected when I chose this project. This leads to the ideas I have for how we should design the workflow for this project.

I wanted to explore using enterprise hardware to deploy a local large language model such as Meta's Llama 2 or its derivatives which holds a relatively lax license for commercial usage. Alternatively, online APIs exist such as the more popular GPT-4 by OpenAI, but I had a meeting with the sponsor back in February, and they have explained to us that LabVoice will be utilized as a cloud solution for processing voice input, automating tasks performed by laboratory equipment, and outputting responses. As of late March, I have been able to fully explore what the software has to offer, and what direction I'd like to take this project.



## Initial Ideas

I'd like to explore using enterprise hardware to deploy a local large language model such as Meta's Llama 2 or its derivatives which holds a relatively lax license for commercial usage. Alternatively, online APIs exist such as the more popular GPT-4 by OpenAI, but we've recently had a meeting with the sponsor, and they have explained to us that LabVoice will be utilized as a cloud solution for processing voice input, automating tasks performed by laboratory equipment, and outputting responses. At this point I am unsure if there will be any hands-on requirements with machine learning libraries, so I will treat LabVoice as an AI black box here.

How I envision the project looking as we progress toward our first prototype is as follows: A user will prompt LabVoice to listen for further instructions with a verbal "wakeword" through an external microphone that is constantly listening for that specific phrase. LabVoice will not respond otherwise. Assuming the wakeword has been said, the user will then instruct it on what task needs to be done. LabVoice will process this speech to text and create a best guess of what the user said. After this, LabVoice will choose the task that most similarly matches what the user requested and ask the user if it should proceed with performing that task. If the user confirms that the task is correct, we will have a pipeline for the data to be sent to a file share which will be accessible to Scitara, a cloud application operating on AWS that will allow us to manipulate any text seamlessly into a specified format such as a JSON or YAML file for example.

We will then create another pipeline for Scitara to send the data to Velocity Labs which will allow us to instruct the laboratory equipment to perform the task that was requested. After the task finishes (or fails), we will extract information from laboratory equipment using Hamilton, which is an

application running on a Windows laptop designed at the lab for this purpose. It will send the raw data through another pipeline back to the file share into a text file. Scitara will process the response and determine whether or not the run of the task failed. If an error occurred, LabVoice should inform the user of the failure and the cause of failure. Otherwise, a notification that the task was successful should be communicated by LabVoice with a potential prompt to request permission with continuing on to the next task that the user has to confirm.

If we want the process to be more automated, I suggest going without this permission prompt and immediately continuing with the next task in the workflow unless manual intervention is required before moving forward. This process could continue iteratively until the end of the requested workflow has been reached. To implement this, we would need to figure out if we can get LabVoice to stack requested tasks in a queue with verbal prompting alone ideally. Once the final task has been run, LabVoice should generate a different notification phrase to alert the user that it is finished. This is a high level example as I am not very familiar with many of the tools being used here, but I would like if the general workflow worked like so. Additional stretch goal ideas I have are things like creating a GUI interface that users can interact with as a backup in a scenario where LabVoice cannot be used. This could happen in a scenario where a user who is hard of hearing or even deaf needs to use the laboratory equipment. It would allow them to at least schedule tasks to be run in succession and work nearly as efficiently as any other user working in the laboratory. To take it even a step further, we could create a visual notification that a process has ended with colored lights or something more complex like a mobile push notification.

Huda Shehadeh

Ever since an early age, I have always found artificial intelligence to be super interesting. In my free time, I would watch videos on the topic to help me learn more about its evolving future. When common voice-controlled home assistants like Alexa and Google Home became more popular, I became intrigued by the fact that you could speak to these technologies and make them do tasks like turning on the lights or even ordering groceries to your front door. I wanted to further understand how exactly that was possible and what else these technologies are capable of doing. In my high school years, I decided to participate in a science research symposium competition that required me to choose a topic to research on for a few months and later present to an audience. With my topic being about voice control and its future, I enjoyed taking the time out of my day to learn all about the technology and its processes, including its early development stages back to when the very first voice-controlled device was created. Participating in this event with a topic that I was genuinely interested in made me love every second of it, however, its future outlook sounded even more interesting to me.

Before getting into any college or university, I knew I wanted artificial intelligence to be a part of my studies at some point. After choosing computer science as my major, I definitely feel like it got me closer to having that chance. Considering my previous research and current interest in learning about voice assist technologies, I wanted to be a part of this project to not only further expand my knowledge with the more modern side of artificial intelligence, but to also have the chance to participate in a hands-on project that I can witness the development of. As its future is always evolving, it is exciting to see how artificial intelligence can be incorporated in different ways to help bring more advanced technologies into the world. Being enrolled into an artificial intelligence course has allowed me to learn

more about machine learning and increased my desire to be a part of this senior design project. Knowing that I can have a first person point-of-view experience in working towards its advancements is something I look extremely forward to, and I am beyond grateful to have the opportunity to work with something I have always been passionate about.

## Initial Ideas

As a personal thought, I believe that an effective feature that can be implemented within this technology would be to detect the presence of an unsafe chemical in order to prevent any harm from occurring to users or other people present in the area. In addition, having LabVoice support multiple languages may be extremely useful, considering English may not be a language that everyone is confident speaking in. Considering that there are multiple users who will be working with LabVoice and conducting different tests as compared to one another, it would be ideal for LabVoice to have a customization feature that can adapt to a user's likings and remember them by their name or speak to them in their preferred language. For instance, upon entering the laboratory they can be greeted by LabVoice and it can automatically adjust its settings to fit the user's preferences based on its previous usage. Last but not least, if someone is new to the technology, LabVoice can present the new user with effective tutorials or descriptive directions on how to utilize different devices within the laboratory, including LabVoice itself.

Overall, the goal of this senior design project is to lessen the amount of manual processes used within a lab, by implementing a voice assistant called LabVoice that can be communicated with in helping a user conduct their test successfully. Considering that manual processes can lead to wasted time or even unintentional errors, these new technologies using artificial

intelligence and such can change the environment for the better. As it can help the user feel more confident or relaxed during a test scan, a voice assistant can provide more accurate data that can be transferred with a voice command as a result.

## Caleb Gibson

Personally, I am seeking to do this project for a multitude of reasons. For one, because AI is becoming ever more prevalent in the industry, so developing a solid background in working with various AI technologies – especially those applicable to a multitude of endeavors, such as this “Voice Assistant” technology, which, while mostly software, interacts with other software, which in turn actually controls hardware, to accomplish a task – seems beneficial for a variety of reasons, both vocational and in terms of general skill development. Moreover, I have family and friends who are interested in AI or even have employment working with or in AI, so gaining skills in AI could be beneficial in my relationships with them, even if just for conversational purposes, but perhaps more, such as potentially being able to at least provide them a little bit of help with their jobs if they need or want it. Additionally, one of my best friends was a roommate I had who was a chemistry major and is now going to UCF for graduate school studying radiochemistry, so being able to do a project that has potential applications in an area related to his could be good.

Even more though, now that I have gained a good grasp on what this project is about, I hope to be a part of solving a critical problem in any industry that relies on chemistry or laboratory science and the notion of laboratory science in its entirety. Namely, I hope that I can be a part of the solution to help streamline laboratory work, reducing the time spent by

laboratory scientists doing necessary but tedious tasks, thereby increasing efficiency in the modern laboratory, thereby leading to a host of benefits (hopefully) for all of modern society. I would cover them here, but these benefits and the implications of this project are covered in greater detail in the next section, "Broader Societal Impacts." I hope for the benefits discussed there more so than the potential detrimental consequences, obviously.

Furthermore, this project provides a great opportunity to use modern technology working in a field of interest to me (AI, chemistry, to name a couple) to bring a fascinating and sophisticated industry, with plenty of knowledge and a massive amount of power, but a lack of modern infrastructure (in a few places; they are becoming fewer), into the present day. Making things that solve problems is at the heart of what any engineer does – this project provides an amazing opportunity to do that, with some of the newest and greatest tools.

## Initial Ideas

As far as ideas for the project go, I have a couple. More information was received during the team's second meeting with the sponsor and second senior design team – composed of Industrial Engineer and Data Science students – on Friday, 2/9/2024. This information indicated that the team would likely not be working so much on the artificial intelligence (AI) aspect of the project, such as speech-to-text and textual command processing via generative AI or a large-language model (LLM), unlike what was previously thought, as they would be using an already-existing product to accomplish these parts. This product is known as Lab Voice, comes with a low-code platform for development and integration with scientific instruments' software, and is seemingly provided to Accenture by a third-party vendor,

which will provide more technical information on the tool in time. While the MVP desired by the sponsor is fairly straightforward, I have a couple ideas for how I could contribute to it and also how I could help with development further down the road for elements beyond the scope of the MVP.

For the MVP, I could perform research on the scientific instruments that need to be interacted with for the project and conduct research on how their software works and how to interact with it with software. For example, how to work with the Hamilton instrument's or instruments' software using Python and the Lab Voice assistant's low-code platform, which is probably designed specifically for integration with lab equipment like the Hamilton plate reader. I could also look into how to interact with the Windows File Share system and parse output files from instruments into JSON objects for storage in the database. I prefer to stay away from the side that resembles front-end development and work more on the back-end side, such as working with the database, JavaScript file parsing, Lab Voice, or Python side of the issue.

## **Broader Societal Impacts**

The ASIS Accenture Voice Assistant is making significant strides toward the advancement of technology with the promise of a multitude of positive societal impacts, the most remarkable of that list being the enhancement of medical research speed. Through this project's innovative capabilities, the voice assistant is able to streamline laboratory workflows which in turn revolutionizes how researchers can conduct experiments and analyze data. With one of the biggest struggles in a lab being having to switch between softwares and different instruments that don't communicate well with one another, this feature alone dramatically expedited the process within the lab environment eliminating that issue.

Imagine being a researcher faced with a complex experiment and having to switch between multiple software platforms, manually adjusting instrument settings, and meticulously recording the data at each step of the process. With ASIS Accenture Voice Assistant, these complexities can be reduced to simple voice commands. This allows for fluid experimentation with transitions between tasks occurring seamlessly and having minimal interruptions. Not only does it accelerate the pace of research, but also frees up valuable time and cognitive resources to allow researchers to focus more on the scientific research at hand. Furthermore, the ASIS Accenture Voice Assistant offers a user-friendly interface that empowers researchers of all backgrounds to engage with cutting-edge tools and methodologies. It allows for inclusivity and fosters greater collaboration within the scientific community and opens doors to new perspectives and approaches to innovation.



Beyond the ASIS Accenture Voice Assistant's role in accelerating medical research speed, since it serves as a central hub for all laboratory processes, it facilitates more accurate and rapid data production. Within the typical environment of the Accenture lab, data usually traverses through several software systems such as Scitara and LabVantage LIMS, each requiring meticulous attention and manual intervention. With the integration of our technology, researchers are freed from the burdensome task of manually navigating through these systems.

In lieu of expending valuable time and energy on tedious data processing tasks, researchers can now rely on the ASIS Accenture Voice Assistant to handle these time consuming processes. By providing a unified platform that homes the interconnectivity of all essential software systems, our technology streamlines data production, minimizes errors, and maximizes efficiency. Researchers no longer have to waste valuable time doing everything manually. Instead, they can entrust the responsibility to the voice assistant, freeing themselves to focus on other productive activities during the processing time.

The significance of ASIS Accenture Voice Assistant becomes particularly pronounced in domains such as Pharmacology, Chemistry, and Biochemistry, where research processes demand precision and accuracy. If there is ever any slightest deviation or error in data production, it could have extreme implications that could impact the development of life-saving medications or therapies. For instance, in Pharmacology, the efficacy and safety of pharmaceuticals rely upon the accuracy of data generated during the research phase. Minor inaccuracies in things like dosage calculations could ruin the integrity of clinical trials and delay the delivery time of essential treatments to patients. It is a similar case for Chemistry and

Biochemistry in the sense that researchers explore chemical reactions and biological processes where precision is non-negotiable. A single miscalculation or oversight could lead to conclusions that are way off which in turn hinders scientific progress and innovation.

Moreover, the urgency of discoveries in these research fields cannot be overlooked. Since time is of the essence, with every moment saved in the research process it translates to potential lives improved or even saved. The ASIS Accenture Voice Assistant's ability to expedite data production and streamline workflows becomes a critical asset that enables researchers to accelerate the pace of discovery and innovation.

The impacts highlighted above are not the only positive changes that the ASIS Accenture Lab Assistant brings about, it also brings to light the issues with all these laboratory instrument companies creating instruments that don't communicate well with each other. By addressing the problem and working towards solving it, it brings change to the laboratory research fields.

## **Legal, Ethical, and Privacy Issues**

While this project itself has not been and is likely not subject to any legal, ethical, or privacy concerns, the field and industries to which the results or the solution (assuming that one is developed) could be applied are certainly ones where there could be legal, ethical, or even privacy issues. As far as this project itself and the direct ramifications of a hypothetical successful solution to the sponsor's problem through the completion of this project, the only immediate conceivable concerns are that the successful automation of parts of – or even (if the team is somehow very successful) the entirety of – the modern laboratory workflow and processes could lead to job losses as a result of cost cutting and improved efficiency – a starkly corporate and cold take on the possibilities that such technology could have for humanity, but one that is nevertheless almost sure to be realized by at least one company that incorporates such a solution into their workflow.

Beyond the direct ramifications of the team's successful completion of the project, however, lies the amalgamation of legal, ethical, and privacy issues that such a technology could have in the industries to which it is applied. For example, this laboratory-process-streamlining project solution could be used to great effect to improve the data collection rates and drug testing rates of the pharmaceutical industry. However, if implemented without proper safety precautions, it could have devastating effects, like leading to an uncaught error in a chemical reaction or in a pipetted sample volume, inflicting misleading results that could have terrible repercussions if a drug based on such uncaught faulty readings was to receive regulator approval and be served and marketed to the masses, or even simply to the initial trial group on which it is tested before receiving regulator consent for sale to the mass consumer markets.

For privacy concerns, it should be noted that a very common laboratory task – according to both the sponsor’s team members, all of whom have a scientific or technical background, and the LabVoice team – for which this technology could be (and even would likely) be used is for handling, performing measurements regarding, and storing said measurement data regarding DNA samples, blood samples, or other biomedical materials – all of which, if human, could identify or otherwise be medically related to the individual they are sourced from. This is not our concern, however it would most definitely be a worry if a client using the technology solution that the team creates employs it for that kind of role, so care must be taken to ensure data accuracy should the team go the route of storing any data related to a potentially sensitive sample type (that is, sensitive with regards to the data that can be obtained or measured from it).

# **Project Characterization**

## **Objectives and Goals**

This project has several objectives and goals. The sponsor – a team of scientists, mostly from a chemistry background, but some from a technical background, working for Accenture to develop a product asked for by a client – wants this project to accomplish a few main goals for its minimum viable product<sup>5</sup>. These goals are encompassed by a voice assistant which can perform the following four core functions:

- 1) First of all, the LabVoice voice assistant should be able to turn a specified laboratory instrument on.
- 2) Second of all, the LabVoice voice assistant should be able to run said instrument.
- 3) Thirdly, the voice assistant should be able to report when that instrument has encountered an error if that instrument is running.
- 4) Fourthly and finally, the voice assistant should be able to report when that instrument is done running.

Beyond the MVP, there are other goals and objectives that the team wishes to achieve. The most significant of these is to implement, integrate, and verify (test) the voice assistant developed by the team such that – should it be appropriate for the instrument and operation in context – the

---

<sup>5</sup> MVP

output of an instrument is captured, stored in a file, and both forwarded to and stored on the sponsor's bundled LIMS/ELN for the possibility of remote access, as well as stored locally on the machine to which the file is initially sent or interacted with to obtain the file from.

As a stretch goal, the sponsor has requested that an artificial intelligence layer be implemented to help streamline the workflow and improve the accuracy of voice input received in LabVoice. Eventually the sponsor would like to have data produced by their laboratory equipment be analyzed and parsed by an AI into a more user readable format as it currently is all pushed into a simple text (\*.txt) file in a very simple comma separated format. One proposed solution was implementing a large language model (LLM) and/or discrete natural language processing (NLP) program that could be responsible for this. During research and investigation, it was found that implementing a discrete NLP solution was less effective and much more difficult to create, so a LLM agent was selected to perform both responsibilities.

# Specifications and Requirements

## Minimum Viable Product – High-Level Description

The sponsor has thus far specified that an MVP for the project is comprised of an integrated voice assistant using LabVoice – developed utilizing the LabVoice low-code platform, which is web-hosted and can be accessed through either a web application or a mobile application – that can be invoked by voice to perform a preset sequence of events. This preset sequence of events is more specifically to be a laboratory workflow. The sponsor has demonstrated multiple laboratory workflows that they have that the team may select from. These range from quite simple to extremely intricate.

Regardless of the laboratory workflow utilized, for the MVP, this preset sequence of events can be summarized as follows.

- 1) First, the voice assistant should be able to be invoked solely by voice, similar to an Amazon Echo device (which is invoked by the user saying, “Alexa,” while within range of the device’s auditory sensors; only in this case, the user may invoke the voice assistant by saying, “Hey LabVoice”).
- 2) Secondly, the voice assistant should be able to interface with at least one instrument or device in the laboratory – and, necessarily, the instrument software on the Windows machine connected to the instrument, directly or indirectly (more on the indirect path later). Specifically, the voice assistant ought to be able to interface with the instrument in such a way that it can command the instrument to carry

out an action. This action could be as simple as powering on (although – depending on the instrument – this may not be possible), or as complicated as running a specific method or sequence on that instrument.

- 3) Thirdly, the voice assistant should be integrated in the laboratory and with the instrument such that the voice assistant or its accompanying application – or another subsystem, such as the instrument itself – can report and emit a notification of a detected error during the action it was commanded to perform, should one occur, to the user.
- 4) Fourthly and finally, the voice assistant should be able to update the user when the instrument has completed whatever action or actions it was commanded to perform as part of the workflow.

This constitutes a minimum for what the sponsor would like to see in terms of the voice assistant's capabilities.

## Extra Objectives – Beyond the Minimum Viable Product

The sponsor has far more ambitious goals for the project (not all of which may be realized fully during implementation and testing). Among the extra goals that will be pursued during implementation, due to these goals' close interwovenness to the minimum goals set forth by the sponsor (alongside their ostensible relative ease of implementation), are the following. Firstly, if the instrument and operation commanded and performed are appropriate for such an additional component of functionality, then an integration should be developed such that once an instrument has completed



an action, the data output by it should be captured and, depending on the desired operation and the action performed, either read out to the user somehow – for instance, displayed on a user’s mobile device screen – or stored to the laboratory’s bundled LIMS and ELN system, or both. It has been mentioned by at least one member of the sponsor’s team that the sponsor has a client that would prefer to see a demonstration of the latter – that is, storing the results in a human-readable format on the sponsor’s bundled LIMS and ELN system – successfully, as opposed to the alternative of storing the results of a connected instrument operation on another cloud hosting service.

## Laboratory Systems and Subsystems

The sponsor’s laboratory contains a variety of systems and subsystems that they desire the LabVoice voice assistant to be integrated with.

### LabVantage LIMS/ELN:



*Image 1. LabVantage logo*

The sponsor’s combined LIMS/ELN product is provided by LabVantage. LabVantage is a very common provider of LIMSs and ELNs in modern chemical and biological laboratories. Although coordinating the LabVoice voice assistant with this LIMS/ELN is not a part of the MVP for the team as outlined by the sponsor, it is a highly coveted extra objective. Ultimately, the files and data output by instruments after taking measurements are sought

to be stored on – and have the possibility of being accessed from – the LabVantage LIMS/ELN.

The LabVantage LIMS/ELN provides a high-level overview and administrator-level control of laboratory processes, tasks, and tools. For the most part, it does not handle exact details so much as it handles overall flows. It is highly customizable and its flow creation platform is designed with users from a non-technical background in mind.



*Image 2. A sample screenshot illustrating some of the capabilities of the sponsor's LabVantage LIMS. Each circle allows the System Administrator – or whatever other LabVantage user – to create or manage something, or to perform some task, related to the functions and work carried out within the lab. Each rounded box, on the left of the screen, represents a category of LabVantage API endpoints.*

To upload and access files, data, and metadata in the LabVantage LIMS/ELN, three major API endpoint categories of the LabVantage LIMS/ELN (see the caption of Image 2) are planned to be utilized as connection points.

These are discussed in more detail in a later subsection, "Design" > "Research and Investigation Process" > "LabVantage Brief Tutorial."

## What is SDMS?

Along with the ELN, we were also demonstrated the SDMS of LabVantage. This is the LabVantage Scientific Data Management System. This robust system provides an overall view of the captured data from various sources, including the instruments, test results, analytics, and research and development initiatives. Moreover, the SDMS provides protection against any potential cyber threats, ensuring the confidentiality of the data assets. By housing critical driver information and facilitating the seamless communication between LabVantage and the instruments, SDMS fosters efficient data management within our project's ecosystem.

## Scitara DLX:



*Image 3. Scitara logo.*

Scitara is a company whose stated purpose is in essence the same as one of the major ends the sponsor is trying to achieve in this project. Scitara aims to facilitate laboratory work and amplify efficiency and speed in modern laboratories by automating data exchange between the disparate instruments, software applications, web services, and informatics systems present in modern laboratories. Scitara seeks to unify them into a cohesive,

easy-to-access, easy-to-use platform, which is their DLX platform. DLX is cloud-hosted (conventionally on AWS), accessible anywhere with an Internet connection, and provides nearly innumerable options to create and customize automated workflows with instruments, systems, and subsystems typical of a contemporary laboratory. Scitara dubs these workflows, “Orchestrations.”

The sponsor currently uses Scitara to largely automate a couple of their workflows. For instance, the sponsor currently uses a Scitara workflow for their BioTek Synergy 2 plate reader. However, they are not voice-activated. They must be triggered manually by a user on their computer. Moreover, while not part of the MVP, they do not upload the files they output to the LabVantage LIMS. Instead, they either retain the file locally or upload it to an intermediary cloud platform provider called Velocity Labs, which stores data on or retrieves it from the AWS cloud.

LabVoice (to Be Integrated):



*Image 4. LabVoice logo*

LabVoice is a proprietary voice assistant product with a low-code development platform centered around creating detailed graphical workflows (represented at a surface level by flowcharts), called “LabFlows”, intended specifically for laboratory usage. These “LabFlows” are composed of “Nodes,” which accomplish certain tasks based on the type of Node, such as “Start LabFlow,” the required Node (meaning it has to be present) that marks the

beginning of the LabFlow, and which can have both parent and child Nodes. The sponsor wishes that the Senior Design team integrate the LabVoice platform with at least one instrument in their chemistry laboratory, and prefer this instrument to be either their main plate reader that they utilize or their automated liquid handler. This will likely be accomplished through a plethora of "JavaScript", "Speak/Listen", "Branch on Boolean Condition", "Branch on Multiple Conditions", and other Nodes, like the "Web service" Node. Each of these Nodes performs a unique function and is highly customizable.

The "JavaScript" Nodes allow developers of LabVoice to write custom JavaScript callbacks, complete with a set of default arguments, several necessary lines of code (such as "let branch;"), and a "try-catch" block, to ensure proper error handling in the event of a script error. As a result, these Nodes are highly customizable and allow for great flexibility and customizability and versatility in development of LabVoice flows and scientific processes with LabVoice.

## Concept of Operations

Users interacting with Voice Assistant will primarily be scientists working at the Accenture laboratory. LabVoice will be the main interface that a user will interact with by utilizing an external microphone. The microphone will always be listening for input from a user, but LabVoice will not accept any verbal commands until the phrase "Hey LabVoice" is said by a user. Once this phrase has been registered by the speech-to-text program in LabVoice, all verbal input will be processed for the user to instruct LabVoice on what LabFlow should be initialized.

After receiving a command from the user, LabVoice will choose the LabFlow that most similarly matches what was requested. These LabFlows are created manually and are extremely versatile in what they can do. For example, if a user requests that the liquid handler LabFlow should be started by saying "Start liquid handler LabFlow", LabVoice will then output a voice prompt requesting that the user reply with the liquid handler method(s) that should be run. The user will then reply with this information, and LabVoice's speech-to-text platform will attempt to interpret the requested methods. If LabVoice fails to find an appropriate match, it will call on a large language model agent to interpret what was captured by the speech-to-text program. LabVoice will then prompt the user to verify if all interpreted methods are correct. If there are any incorrect methods included, the user can cancel the LabFlow by stating "no" or anything else aside from "yes". Otherwise, the user should respond "yes" to prompt LabVoice to send instructions of what methods need to be run to the Hamilton software through Scitara which controls the liquid handler. Once Hamilton receives the instructions, each method will be run in the order that the user verbally prompted LabVoice. If any method fails to properly execute, Hamilton will notify Scitara that an

error occurred, and Scitara will then pass that error over for LabVoice to notify the user.

In the case that no error occurs, LabVoice will inform the user that the LabFlow completed without any issues. From here, the user will likely want to test the samples produced by the liquid handler on the plate reader. To do this, the plate with all the liquid samples on them will need to be transported from the liquid handler to the plate reader for sample analysis. The user will then need to activate LabVoice again by saying "Hey LabVoice" followed by "Start plate reader LabFlow" to begin the plate reader. As the plate reader does not have methods like the liquid handler, there is no need for additional prompting and Scitara will immediately signal the plate reader to begin analyzing the liquid samples. Like the liquid handler though, if any error is detected during the procedure, it will be sent to Scitara and then LabVoice to alert the user. If the procedure completes without error, the plate will eject from the plate reader and LabVoice will inform the user that the analysis was successful. Collected data will then be copied off the local Windows file share and sent to Scitara for cloud storage.

# **Design**

## **Research and Investigation Process**

### Overarching Process

In our research journey, our primary focus has revolved around acquiring training and learning from the experts who are well-versed in various software environments. Through structured training sessions, we were systematically guided through the intricacies of the different software features, gaining practical insights that laid the groundwork for our research endeavors. For example, with LabVoice, we were provided with demonstrations and sample lab flows, enabling us to hone our skills through hands-on practice. This all being an invaluable resource in crafting our own lab workflows.

In addition to the training we received, our team has actively participated in in-person laboratory visits whenever feasible, aiming to acquaint ourselves with the equipment, sponsors, and working environment. These visits offered first hand exposure to various machines, supplemented by personal demonstrations of software functionalities facilitated by knowledgeable sponsors.

To complement these learning experiences, we engaged in remote meetings through Microsoft Teams, collaborating closely with the IE/DS team to remain updated with the sponsor's requirements and address any queries that arose along the way. These interactions served as vital touchpoints, ensuring alignment with project objectives and fostering a collaborative environment conducive to success.



To enhance our understanding and bridge any knowledge gaps, the team has taken a proactive approach by conducting independent research beyond the provided materials. This self-directed learning journey has encompassed a variety of avenues, ranging from simple Google searches to delving into articles and papers touching on relevant topics central to the project's scope. Areas of exploration have spanned an extremely diverse spectrum, including but not limited to generative AI, data processing, JavaScript, and broader discussions on AI itself.

Furthermore, recognizing the importance of grasping the complexities of the tools at our disposal, we delved a little deeper into the manuals of individual lab instruments. This hands-on exploration has provided invaluable insights into the functionalities and operations of these instruments, empowering us with a deeper understanding of their role within the laboratory ecosystem.

By embracing a multifaceted approach to learning, which extends beyond formal training sessions, we've cultivated a rich array of knowledge and expertise. This proactive pursuit of understanding not only improves our development process but also underscores our commitment to delivering a robust and effective solution for the ASIS Accenture Lab Assistant project.

## LabVoice Research Overview: LabVoice API Calling

As part of the team's research towards instrument integration, members of the team arranged and attended at least one meeting per week with the Accenture team and with members of the LabVoice development team, as well as corresponded with them via email extensively.

LabVoice has a number of key Nodes that are present in any or almost any LabFlow. A series of “if” and “else if” statements is replicated by the “Branch on Multiple Conditions” Node, which allows one to define multiple JavaScript conditions, with a branch that is traveled to if a condition is true. The “Branch on Boolean Condition” Node similarly replicates an “if... else” statement. The “Speak” Node causes LabVoice to speak the specified text. The “Speak/Listen” Node allows LabVoice to speak and then listen for a response. The acceptable responses can be extensively customized, as can the actions taken if unacceptable responses are heard or a timeout occurs before input speech is received. Responses are stored in a JSON property “reply” in the “data” JSON object. The “Get Value By Voice” LabFlow is a less customizable version of this Node that has LabVoice speak and then listens specifically for a certain type of value, which it writes to a new JSON property whose name is defined by the Node.

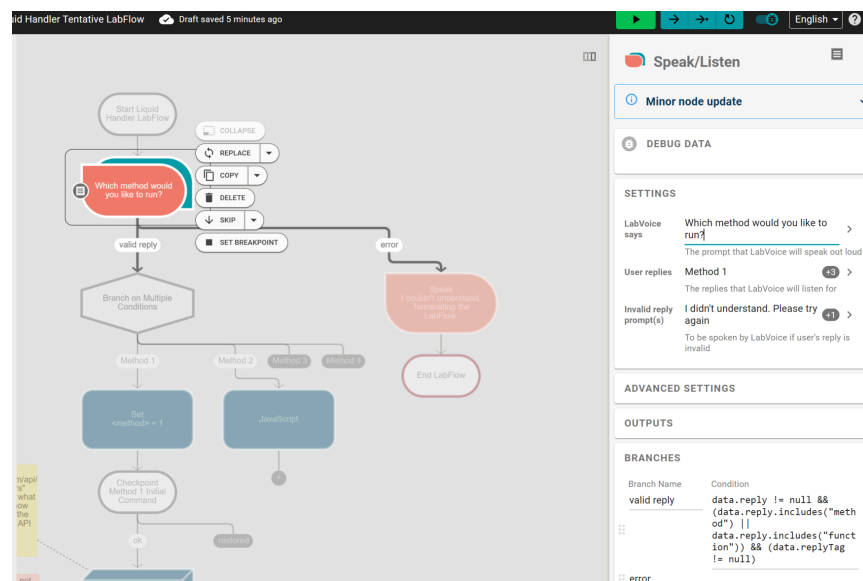


Image 4. A sample LabFlow. This one is incomplete, but still illustrates generally how a LabFlow appears and flows.

LabVoice comes with a Debug Mode that allows developers to view the details of the whole LabFlow, such as the “data” JSON object, and details of

Nodes after running a LabFlow. Debug Mode also comes with a Step Forward feature and a Rerun Step feature. A more scientifically useful LabVoice features is the “Audit Trail” feature, which allows one to keep track of which Nodes were executed and in what order.

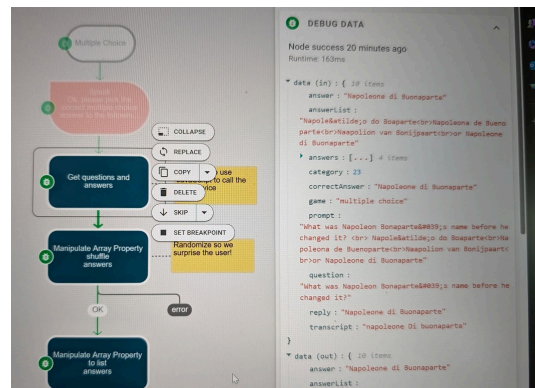


Image 5. A sample of how the "Debug Data" appears after a run of a LabFlow. In this particular LabFlow, a Web Service was used to play a short game of trivia, with the category, "history".

The most important Nodes for LabVoice’s API calling abilities, however, are its JavaScript and WebService Nodes. JavaScript Nodes enable developers to write a pure JavaScript callback using LabVoice’s built-in code editor. Web Service Nodes abstract away some of the API calling details, such as having to program the use of the “fetch()” API versus another API, like Axios or GRPC<sup>6</sup>, by defaulting to using “fetch()” and allowing the LabFlow creator to specify a URL API endpoint with the request parameters and the HTTP method to exercise, while handling OAuth2 (or OAuth 2.0) authorization automatically.

<sup>6</sup> Google Remote Procedure Call

```

1 // Edit this script, replacing the TODO section with your code
2
3 /**
4  * Method handler
5  *
6  * @param {object} settings - read only method settings, matches UI method settings (e.g. settings.url)
7  * @param {object} data - data record
8  * @param {object} globals - read only global values (e.g. globals.timezone)
9  * @param {function(outputs)} callback - callback function to invoke when done, pass data record back
10 */
11 exports.handler = async ({settings, data, locals, globals, callback, modules}) => {
12   let branch; // DO NOT DELETE THIS LINE!!!
13   try {
14     // TODO place your script here e.g. result = settings.parameter1 / 100
15     // TODO add your response to the data object e.g. data.result = result
16     // TODO optionally throw error if needed e.g. throw(Error('toto'))
17
18     // DO NOT MODIFY THE LINES BELOW!!!
19     delete data.error;
20   } catch (error) {
21     data.error = error.message || 'Unknown error';
22   } finally {
23     callback({data, locals, branch});
24   }
25 }
26
27

```

Image 6. Credit: Caleb Gibson. The template JavaScript callback given by the LabVoice JavaScript Node.

To ascertain a better understanding of LabVoice’s capabilities and the capabilities of the instruments that will be interfaced, I have asked numerous questions. For example, I asked how the Script nodes in sample LabFlows that make calls to custom instruments work – this led to me discovering a method of API calling called GRPC, which is more efficient than the typical “fetch()” API calls method partly because its payloads use binary as opposed to JSON or text.

```

1 exports.handler = async ({settings, data, callback, modules}) => {
2   try {
3
4     console.log('settings', JSON.stringify(settings));
5
6     const statusProto = modules.getStatusProto();
7     const response = await modules.send('GetStatus', {'Name': {'value': '
8       [REDACTED]'}, statusProto, settings.host, settings.port, 6000)
9       .catch((err) => { throw (err); });
10
11     if (!response) throw Error('no response');
12     const value = JSON.parse(response.ReturnValue.value);
13
14     data.message = value.Content + value.Error;
15     data.statusCode = value.StatusCode;
16     data.status = value.Status;
17
18     delete data.error;
19   } catch (error) {
20     console.log(error.stack);
21     data.error = error.message || 'Unknown error';
22   } finally {
23     callback({data});
24   }
25 }

```

Image 7. Screenshot credit: Caleb Gibson. Code credit: Unchained Labs. An example of a JavaScript Node using GRPC. The 'value' property in the request is covered for security.

However, it is also more difficult to work with consequently – to how best to deal with errors from the lab instruments to, perhaps most importantly, what specific API endpoints can be used from the sponsor's software tools, specifically what LabVantage LIMS endpoints are available and would need to be used (these have been determined to be the “Instruments”, “Methods”, and SDMS LabVantage menu trains), what Hamilton liquid handler endpoints are available and are most likely to be employed, what Scitara API endpoints are available and how they could be used to interface with LabVoice such that we do not have to put in as much effort to start a run on the instrument, retrieve the data from it, and upload that data to LabVantage using LabVoice, and finally what Velocity Labs endpoints are available. These last two questions were only asked by one of my other UCF CS Senior Design team members, Christopher Fong, or Chris, and I, after we jointly attended a meeting with the sponsor during Spring Break to comprehend Velocity Labs better. Namely, how Velocity Labs and Scitara fit into the sponsor’s existing semi-automated workflow with the Hamilton liquid handler, what exactly the Velocity Labs data flow looked like, and how and whether we could integrate LabVoice with it or Scitara to accomplish the task of running an instrument by voice more easily than we had initially planned.

Ultimately, interacting with the API endpoints or the various software tools that the sponsor desires us to utilize will require more research. It will require both more meetings with the sponsor and all the various development teams that they are working with, as well as other digital and email correspondence with the makers and vendors of those various products and software tools.

## Initial Laboratory Visit

Our initial visit to the laboratory served as a pivotal moment in our project journey, affording us the opportunity to interface with the project sponsor and gain a comprehensive understanding of the Accenture ASIS Lab Assistant project's overarching goals and scope. With administrative matters efficiently addressed, our attention turned towards acquainting ourselves with the diverse array of instruments housed within the laboratory premises.

Situated within the Accenture office laboratory, there are no high-risk experiments necessitating hazardous chemicals, meaning that the instrumentation presented is characterized by simplicity and practicality. At the forefront of the entire ensemble stood the Liquid Handler or Auto-pipette, which is a state-of-the-art apparatus adept at precisely transferring reagents into a 96-well plate. This cutting-edge machinery was seemingly the most modern piece of technology among its counterparts. Complementing this device were other instruments such as the centrifuge and thermocycler, seamlessly accommodating the 96-well plate. Across the laboratory housed the essential tools like balances, scales, and plate readers, each playing a pivotal role in facilitating the research needs and operations.

Following informative demonstrations, we delved into the underlying challenges and obstacles prevalent within the laboratory environment. A primary concern of the equipment originated in the fact that the instruments are sourced from various manufacturers, causing an obstacle to seamless interconnectivity and communication. This issue is our team's principal objective, as we are trying to establish a cohesive environment conducive to streamlined operations.

Furthermore, our visit provided valuable insights into the intricacies of data management, as we perused through various CSV files, JavaScript documents, and other data-related artifacts. This wide range of foundational knowledge helped the team navigate the complexities of data processing and analysis. In essence, our initial laboratory visit served as a cornerstone, laying the groundwork for our subsequent research endeavors. With these newfound insights our team had a clearer understanding of the project objectives and were more prepared to drive forward with the innovation within the Accenture ASIS Lab Assistant project.

## LabVoice Demonstrations

The team's first introduction to the LabVoice platform came on Friday, March 1, 2024. On this day, the team and two members of the sponsor's team, Mr. Kelliher and Mr. Ellis, attended a one-and-a-half-hour-long virtual presentation and tutorial of LabVoice and its capabilities demoed by some members of the LabVoice team, including but not limited to Mr. Tobias Rothe, Mr. Dave Rogers, and Mr. Frederic Bost. Mr. Rothe has been the team's most frequently contacted point of contact for asking and getting answered technical questions about LabVoice.

## Manual Pipetting Laboratory Workflow Demonstration

This demonstration took place during the same meeting where one of the team members went to set up a remote desktop connection to the automated liquid handler-connected laptop (discussed in further detail in the next subsection). This occurred on March 14, 2024, in the sponsor's ASIS Orlando laboratory. The other senior design team that the sponsor is working

with, the Industrial Engineering and Data Science senior design team, was present. The sponsor was performing time trials for them, demonstrating how long it takes to manually complete the sponsor's demo workflow of:

- 1) manually pipetting samples into a 96-well plate, then
- 2) inserting them into the BioTek Synergy 2 plate reader, then
- 3) going into the plate reader's companion software app on the Windows laptop connected to it, then
- 4) running the basic absorbance reading function with the Windows plate reader app software, then finally
- 5) receiving the absorbance results matrix for the 96 wells of the plate and storing it (in this case, it was not uploaded to Scitara, parsed, and sent back in a text file with the cleaned and processed data).

This workflow took over five minutes to complete, on average (Mr. Kelliher completed it three times for the IE and DS<sup>7</sup> team), and required active human participation the entire time. This was contrasted with the workflow that relied on the automated liquid handler, which, while also taking approximately five minutes to complete, did not require human intervention until the automated liquid handler had finished pipetting the samples into the 96-well plate, more than three minutes in. While this was only for one 96-well plate, it is easy to see how with this kind of operation scaled up, with multiple plates, and possibly even multiple plate readers, automation could provide a gateway to far better realization of human potential in the

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<sup>7</sup> Industrial Engineering and Data Science



laboratory and to progress towards increasing research, production, and testing speed.

Along with an in-person tour of all the machinery, our sponsor provided instructions on how to get the Hamilton Liquid Handler started in the lab. This list being:

- 1) Turn on the liquid handler.
- 2) Open laptop connected and login.
- 3) The Hamilton Method Editor app is on the desktop. Once open, open the Dilution method.
- 4) Open Run Control from Hamilton Method Editor
- 5) The Run Control window will pop up. It will attempt to connect to the instrument right away
- 6) Click the play button to run the method
- 7) The first step will prompt to select the status of the tip box. Assuming it's full, all tips should be selected. The liquid handler should know which tips are remaining.

This list of instructions will be immensely useful once I get to the implementation stage of the project. These outlined instructions that they provided is what we would use as guidance to work with and automate with LabVoice. With the LabVoice component setup, we aim to simplify and expedite laboratory operations for a smoother and more efficient workflow for the researchers.

## LabVoice Integration: Remote Desktop Option for Development and Integration

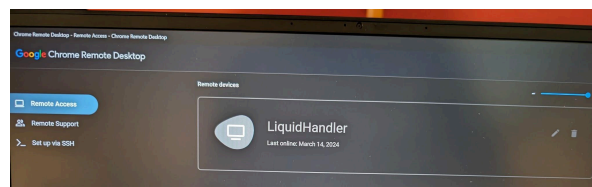
Until a pair of Friday meetings with the sponsor where Mr. Angelelli was in attendance on March 29 and April 5, 2024, efforts had largely been focused on instrument integration and researching how LabVoice can be integrated with the two more complicated instruments, the plate reader and the automated liquid handler. Beginning in March, of the two instruments, these research and development efforts had mostly been inclined towards the automated liquid handler. In order to access the automated liquid handler's companion Windows application, VENUS, one must have access to the computer connected to the liquid handler. This means one must either be physically present in the laboratory with the computer – which schedule conflicts had mostly prohibited, as discussed in the "Setbacks" section – or have some way to remotely access it.

To this end, one of the team members researched options for a remote desktop connection. By meeting with Mr. Rothe and doing Internet research, in addition to trial and error while attempting to set up the remote desktop connection, that team member ultimately settled on Chrome Remote Desktop as the remote desktop connection solution of choice. That team member received approval from his team and from Mr. Kelliher before doing so.

Chrome Remote Desktop was selected for its relative ease of use and security opposed to the two other options, both of which relied on the built-in Windows' Remote Desktop Connection service on the laptop connected to the liquid handler. To set up the remote desktop connection, the team member traveled to the sponsor's lab in Kissimmee. Mr. Kelliher did

not have easy access to a VPN, so setting up a secure Remote Desktop connection such that the liquid handler laptop would be available outside the Accenture office's LAN without being insecure was not straightforward with Remote Desktop Connection. A solution using port forwarding could have been implemented, but this would expose the liquid handler's connected laptop to Internet access for any actor, making it not secure. Additionally, while the information necessary for a port forwarding solution, such as the router IP address, could have been obtained, it would have been difficult to do so in the time allotted and with the Accenture staff available in their office at the time.

After Internet research and experimentation to discover how to configure it so that every member of the UCF CS team could use Chrome Remote Desktop to access the liquid handler, that team member created a shared group Gmail account and logged into Chrome Remote Desktop on the sponsor's liquid handler laptop with it. Then, he connected his personal laptop to a mobile hotspot on his phone and logged into Chrome Remote Desktop with that shared Gmail account, to test accessing the remote desktop from outside the Accenture office's LAN. It worked, and with some guidance from the sponsor that team member was able to initiate a run of the liquid handler with the VENUS software from my own laptop and observe the liquid handler working as expected.



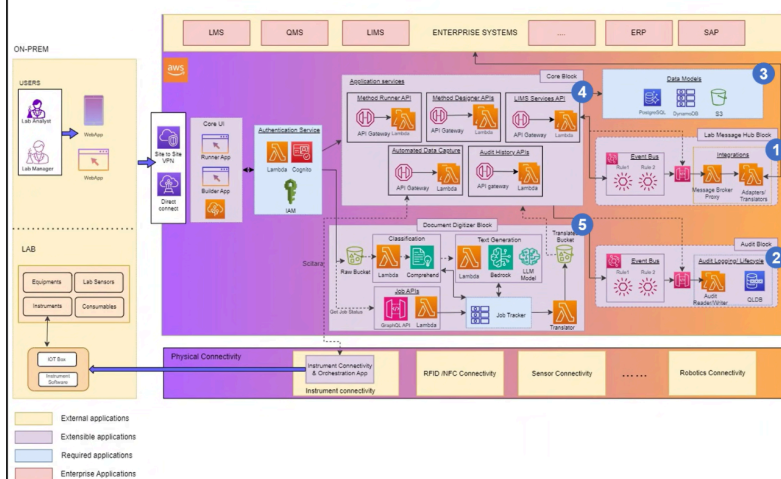
*Image 8. The Remote Desktop option now that it is configured. The liquid handler machine has been named "LiquidHandler" for convenience.*

## Velocity Labs Workflows Tutorial and Demonstration

This meeting took place during Spring Break on Thursday, March 21, 2024, with several members of the sponsor's team and CS SD team present. It was spearheaded by Mr. Shabbir Kheraluwala, a technical expert on the sponsor's team. It was also attended by Mr. Mitchell See and Mr. Michael Kelliher, on the sponsor's team. This meeting highlighted one of the sponsor's pre-existing workflows that they had set up for their Hamilton Microlab NIMBUS automated liquid handler. This particular workflow made use of something called Velocity Labs, which is a proprietary vertically integrated platform for laboratory solutions that provides a broad range of services for orchestrating, facilitating, and streamlining laboratory processes and workflows, and even for performing some of the tasks a LIMS or ELN can perform (albeit absolutely far from all of the functionality that a LIMS or ELN can perform, as they can perform very many functions, beyond what most people could even conceive of by themselves).

During this demo, Mr. Shabbir Kheraluwala exemplified much of the functionality available through the sponsor's Velocity Labs workflow. During this workflow it was shown that the necessary configurations for a remotely started run of the "Dilution" method as a test with the automated liquid handler included many factors. For example the locations of the wells to deposit liquid samples in, given by their row letter and column number, the rack they were located on based on the rack's number, the position of the rack on the plate reader, for instance closest to the automated liquid handler's robotic pipetting arms base or offset a certain rack count, and the rack's orientation on the plate reader rack conveyor section, meaning oriented lengthwise along the rack or widthwise.

## Architecture that is focused on laboratory operations but integrated for end-to-end enterprise processes



Digital Lab Platform is comprised of...

- 1 Enterprise Integration Services** to automate data flow between lab systems and store data into data structures
- 2 Cross-Application Audit Services** to capture, store, and track audit data in an immutable, ledger-based data store
- 3 Data Models** designed for labs to store structure & unstructured lab data with audit trail and full traceability
- 4 Lab Orchestration Services** to drive workflow-based processes, connect lab instruments, automate test run and collect data
- 5 Method Ingestion Services** to ingest & digitize structured documents/SOPs using NLP & GenAI technologies

Image 9. Block diagram illustrating the connections between various components that the sponsor envisions as part of automating laboratory workflows and processes. Specifically, this diagram depicts such an automation concept that uses Velocity Labs, which provides a vertical platform for workflow triggering, execution, and subsequent cloud storage on the AWS cloud of results, data, and metadata (needed for maintaining audit trails).

## Instrument Connectivity via Scitara

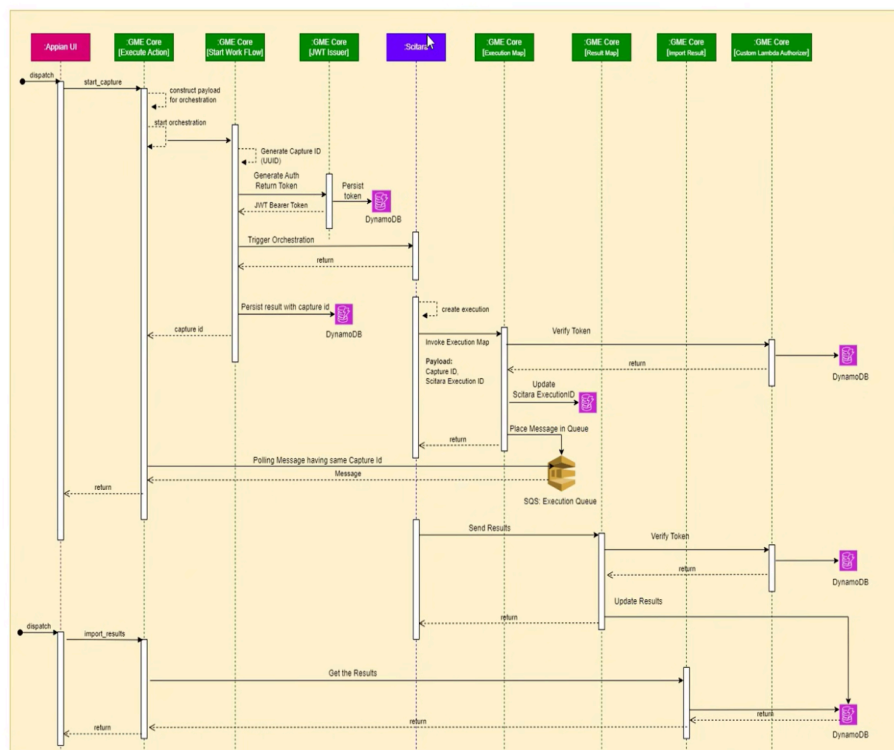


Image 10. Sequence diagram illustrating connections between various components, including Scitara and Velocity Labs's AWS DynamoDB database, as part of an automation of automated liquid handler workflows and processes. The end-to-end workflow of a Velocity Labs-triggered workflow, demonstrated for a Hamilton automated liquid handler like the one in the sponsor's ASIS Orlando laboratory, is shown. Contrary to the sponsor's MVP, this workflow requires clicking of a button on an app with an Appian UI, and also does not deliver data for storage to the LabVantage LIMS, instead relying on the Velocity Labs platform.

## LabVantage Brief Tutorial

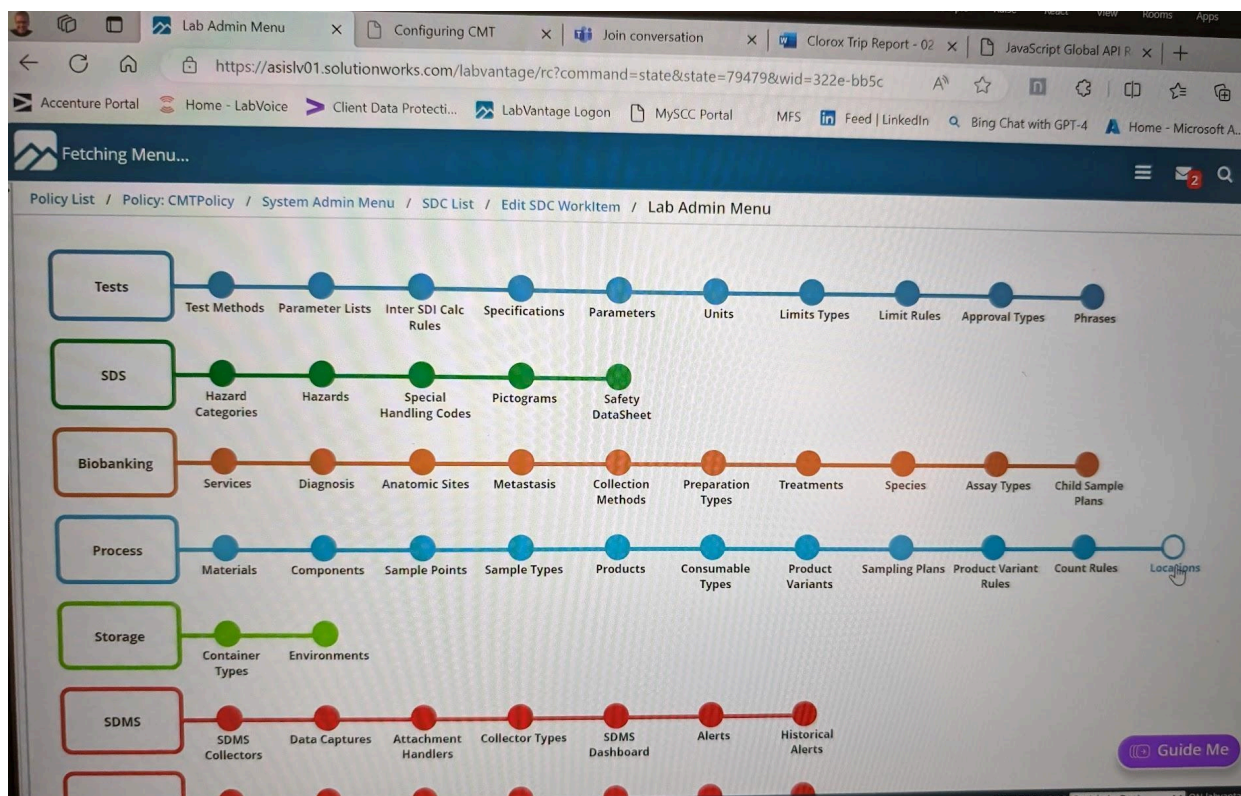


Image 11. A view of some of the menu point categories or connection point categories available in the LabVantage LIMS. Per Mr. Robert Ellis, the three that will probably be necessary for the team to work with for data upload (JSON, txt documents, etc.) are Instruments, SDMS, and Methods. Only SDMS is pictured here.



Image 12. A zoomed-in perspective of the "SDMS" connection point or menu train category in the LabVantage LIMS. As Mr. Robert Ellis stated during one of the team's meetings with him regarding LabVantage, this "SDMS" is to be one of the three major connection points that the team will have to use in connecting LabVantage to Scitara. SDMS is what allows the communication between LabVantage and the instruments to happen, according Mr. Rob Ellis.

Mr. Rob Ellis gave the UCF CS SD team a demo of LabVantage on Thursday, March 21, 2024. During this demo, he expounded a few of the many features of LabVantage, and provided a list of three primary “menu trains,” which are essentially connection point categories, that the UCF CS SD team will likely have to work with. These three menu trains are:

- 1) Instruments – which refers to the types, makes, models, port numbers, ID numbers, and other distinguishing factors of the instruments in the lab that are hooked up to the LIMS,
- 2) Methods – which refers to the methods that each of these associated instruments can perform and specifically how they as methods are intertwined with the LIMS (for example in terms of uploading data and downloading data or execution IDs or sending audit trails or what have you),
- 3) SDMS – meaning Scientific Data Management System, which refers to methods to handle the data communication between instruments and the LIMS in and of itself.

Information on samples can be pulled from LabVantage using a guided method execution, where methods can run and use Scitara for data mapping. Considering each sample gets assigned a unique ID number, they can be sent through Scitara using this high-level workflow to deliver any type of important information or metadata, including plate positions, expected result attributes, testing theories, or others that are worthy of noting. Data can also be transcribed onto a document simply with a voice command, where results can be sent back to a software like LabVantage once they are ready to be published.



Since laboratory tests can consist of many responsibilities and tasks that are often split among a team of people, it is important to note that LIMS can dynamically assign these tasks in order to prioritize deadlines and urgent samples/experiments. Often times these softwares collaborate with other software solutions like Electronic Laboratory Notebooks (ELN) work together to deliver an enhanced data management system by focusing on scientific innovation. For instance, ELN Material Weighing focuses on speeding up the slow process of manual sample collection methods that may have led to human transcription/memory issues.

The ELN Material Weighing process allows scientists or other users to use a virtual assistant to check for important information on a reagent, where an end-to-end integration to the ELN and Inventory System is able to confirm and automatically update data.

To further explain its process and relate it to its uses within the project's senior design laboratory, the ELN in question would be accessed by the user in order to fill in materials and their expected weights within the lab. After visually confirming the correct substance for the tests, the user will then keep note of expected results before moving on to get confirmation. Data like the substance's weight, container ID, or lot number may be transcribed into an ELN, where the same process for different reagents will repeat.

It is important to note that within manual labor, it is common for people to take breaks or stop something they have been continuously working on to deliver a different task. In the case where they wanted a picture of something emailed to them during a test, unintentional errors in future testing may occur considering this break that would have to be taken.

For example, stopping the test to grab a photographic device, adjusting the camera or lighting to get the perfect picture, and even sending the email may not sound like a whole lot of steps but is definitely enough to distract one from the tasks they were previously doing. Since LabVoice is capable of doing a command like so, it is very beneficial to take such for granted.

## Scitara Tutorials and Orchestration Demonstrations

### Scitara Tutorial

This tutorial took place on Friday, March 29, 2024. It was primarily motivated and guided by Mr. Chris Angelelli, a more technically oriented member of the sponsor's team.

### Second Scitara Tutorial

This tutorial took place in-person in the sponsor's ASIS Orlando laboratory on Friday, April 5, 2024. Similar to the "Scitara Tutorial" meeting the previous Friday, this meeting was primarily motivated and guided by Mr. Chris Angelelli, a more technically oriented member of the sponsor's team. In fact, he was the only member of the sponsor's team in the lab with the UCF CS SD team.

## JavaScript Research

At the beginning of the project, one thing that the sponsors emphasized the importance of was JavaScript. They noted that JavaScript was going to be used for data processing among other things. Motivated by

this directive, we took it upon ourselves to brush up on our JavaScript skills. We first went through W3Schools and learned the basic syntax of everything because a good foundation is the key to success in anything. Given certain team members' lack of expertise in anything involving Artificial Intelligence, we also decided to look into how JavaScript can be applied for voice command technology. At this time we did not know that we had a software built to deal with voice commands but we believe our research was useful knowledge to have anyway.

We found that there were already speech recognition objects and event handlers available that could be implemented into our code. Another way to do this is to also leverage speech recognition APIs for commands thereby eliminating the need for continually appending multiple commands directly into the codebase. Through further investigation, We also discovered Tensorflow.js, a widely recognized tool which is commonly used in machine learning and can be used to implement speech command recognition. With Tensorflow, you can use transfer learning and a pre-trained model and adjust it depending on your voice command dataset.

## Large Language Model Agent Research and Testing Results

LLMs are deep learning algorithms that can perform a variety of NLP (natural language processing) tasks. In practice, many people refer to it as generative AI or even more widely known as ChatGPT by OpenAI which is one of the most well-known pioneers of the technology. Alternatives such as Google's Gemini, Mistral AI's Mixtral, and Meta's Llama 2 are all competing models with different usage licenses. For our research, we started off with Llama 2 because of its relatively free license and open-source nature. Meanwhile, OpenAI's best model, GPT-4, and Google's Gemini Ultra may be superior in performance, but have much more restrictive licenses which disallow for commercial usage, and they are closed-source black box programs.

Initially, we planned for this to be much cheaper since we would have been paying for API usage if we chose to use a closed-source model. We experimented with smaller sized models that could fit in a standard consumer desktop computer. The smallest sized Llama 2 models are 7 billion parameters which can fit in most mid-range gaming computers, but these models ramp up to 13 billion, and finally 70 billion which would have issues even in high end desktops. As we began implementing the LLM into our system, we quickly realized that the demands of running such large models were far beyond what our current hardware could handle efficiently. The original setup consisted of a NVIDIA GeForce RTX 3070 graphics card with 8 GB of video memory (VRAM), which had served well until now. However, it quickly became clear to us that running a competent model locally required something more powerful.

We began researching and familiarizing ourselves with all the technologies necessary to simulate how VoiceAssist would utilize them such

as large language models described above and speech-to-text processing which we expected to use for the project if time allowed. More specifically speaking, we looked at using different fine-tuned versions of Llama 2 and Retrieval-Augmented Generation (RAG) to see how feasible it would be to have generative AI analyze human requests or even laboratory data and format it in a consistent manner. For example, we found <https://huggingface.co/jondurbin/airoboros-m-7b-3.1.2>, a 7 billion parameter LLM created by Mistral and fine-tuned by a third party which can take the input "Create a MathJSON solution to the following: Calculate the area of a circle with radius of 17.2456cm." and format it into a MathJSON format that we could have interpreted by another program. The output from the LLM returned as the following:

```
<mathjson>
[
  "Multiply",
  "Pi",
  [
    "Power",
    17.2456,
    2
  ]
]
</mathjson>
```

One idea that arose from this was potentially having the LLM interpret speech-to-text input from a user and format it in such a way that will always select a procedure or multiple procedures on the lab equipment given a certain phrase is in the prompt such as "Start..." X procedure.

We began by setting up a locally hosted 7B LLM from Mistral after comparing benchmarks that showed it was superior to Llama 2 7B and connected it with Whisper speech-to-text from OpenAI which would simulate the responsibility of LabVoice's speech-to-text that we did not have access to when this was put together. We managed to connect the two so that it could hold a back-and-forth conversation using the LLM with solely voice input as LabVoice would operate. For a few weeks, we had been extensively testing the capabilities of the 7 billion parameter model. It was a fine-tuned version of Mistral AI's 7B model that we found to be quite disappointing as it often was confused and generated responses differently than we wanted it to. The model size limitations were frustrating, so we researched better choices, and in the end, we narrowed it down to running two NVIDIA GeForce RTX 3090s that each have 24 GB of VRAM.



*Image 13. Credit: Christopher Fong. The final computer which had two NVIDIA GeForce RTX 3090s that split the loaded LLM into the VRAM of each GPU.*

Loading the model on regular system memory, or more commonly referred to as RAM, was another option we considered, but we found it way

too slow compared to loading it into a graphics card's memory. With this, we were able to run much "smarter" LLMs on the system that would hallucinate far less due to their larger datasets that they were trained on.

### Inference-time Parameters

We had also researched the impact of different inference-time parameters that are provided when prompting the model. These included, temperature, dynamic\_temperature, quadratic\_sampling, top\_k, top\_p, typical\_p, epsilon\_cutoff, eta\_cutoff, token frequency sampling, top\_a, min\_p, and mirostat. Each of these parameters can greatly influence the vocabulary, quality, and randomness of the reply received from the LLM, so we dove deep on the impact of each one.

**Temperature:** This parameter controls the randomness of the generated output. A higher temperature (e.g., above 1.0) leads to more diverse and creative responses, while a lower temperature (e.g., close to 0.0) produces more deterministic and focused outputs.

**Dynamic Temperature:** Similar to the regular temperature, dynamic temperature adjusts the randomness during text generation. However, it adapts dynamically based on the context or position within the generated sequence.

**Quadratic Sampling:** This technique influences how tokens are sampled during generation. It allows for more control over the probabilities assigned to different tokens, potentially improving the quality of generated text.

**Top-k:** Also known as "nucleus sampling," top-k restricts the vocabulary during text generation to the top-k most likely tokens. It ensures that only a subset of tokens is considered, enhancing the

coherence of the output.

**Top-p:** Similar to top-k, top-p (or nucleus sampling) limits the vocabulary to the most probable tokens, but instead of a fixed number, it considers tokens whose cumulative probability exceeds a certain threshold ( $p$ ).

**Typical-p:** This parameter defines the typical value of  $p$  used in top-p sampling. It influences the balance between creativity and coherence in the generated text.

**Epsilon Cutoff:** Epsilon is a small positive value added to probabilities during sampling. The epsilon cutoff determines when to stop considering tokens based on their probabilities. It prevents overly rare tokens from being included.

**Eta Cutoff:** Similar to epsilon cutoff, eta cutoff determines when to stop considering tokens based on their probabilities. It helps control the trade-off between diversity and coherence.

**Token Frequency Scaling (TFS):** TFS adjusts the token probabilities based on their frequency in the training data. Rare tokens may be scaled up or down to influence their likelihood during generation.

**Top-a:** An alternative to top-k and top-p, top-a selects tokens whose cumulative probability exceeds a certain proportion ( $a$ ) of the total probability mass. It dynamically adapts to the token distribution.

**Min-p:** Min-p sets a minimum probability threshold for tokens during sampling. Tokens with probabilities below this threshold are excluded from consideration.

**Mirostat:** Mirostat is a novel approach that combines various sampling techniques to achieve a balance between creativity,



coherence, and diversity in generated text.

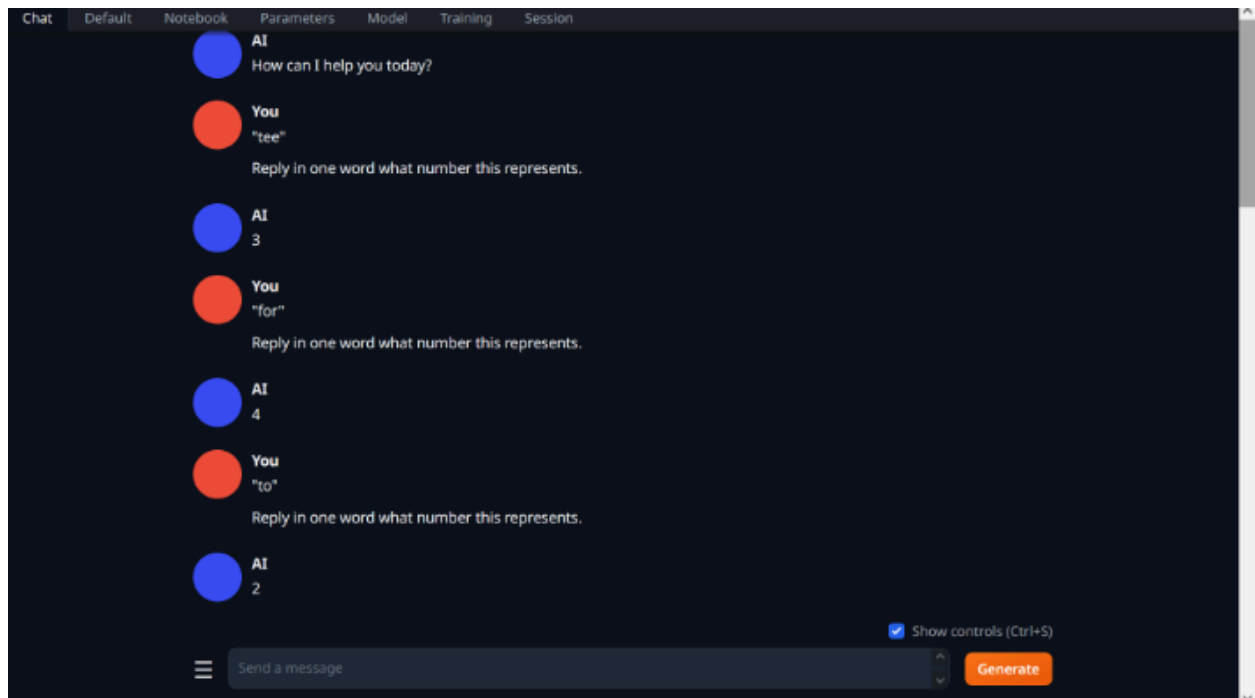
As we experimented with the different parameter settings, we learned that increasing the randomness and probability of the LLM generating tokens that are normally excluded from consideration would ruin consistency in replies. Temperature and min-p would play a significant role for these. For example, if we kept the temperature parameter as a high number, the likelihood of receiving a similar answer to an open-ended question would be much lower. Adjusting these settings, we were able to utilize the LLM so it would consistently respond the way we wanted it to. Considering how a liquid handler LabFlow was expected to run, we wanted this LLM to be able to capture any instances where LabVoice's speech-to-text misinterpreted what method the user requested as a non-numeric input.



*Image 14. Credit: Christopher Fong. An example of LabVoice incorrectly capturing the verbal input "run procedure 2" as "run procedure to". This was a consistently replicable issue that occurred when attempting to capture this input.*

After repeated trial and error, we were finally able to prompt a fine-tuned variation of Llama 2 to respond in a consistent manner. This can be hosted on an AWS EC2 instance as proposed by the sponsor, and it could

possibly be recreated with an even more powerful model that goes far beyond what consumer hardware can handle. Further research into RAG and newer models that have recently been released may allow for more advanced implementations such as processing results output by the plate reader instrument. This would require a large sample of previous output samples from the plate reader to minimize hallucinations that commonly occur with LLMs that are under-trained on a niche dataset.



*Image 15. Credit: Christopher Fong. An example of our locally hosted Llama 2 LLM agent correctly interpreting the intended number from a speech-to-text misinterpretation with no additional prompting context.*

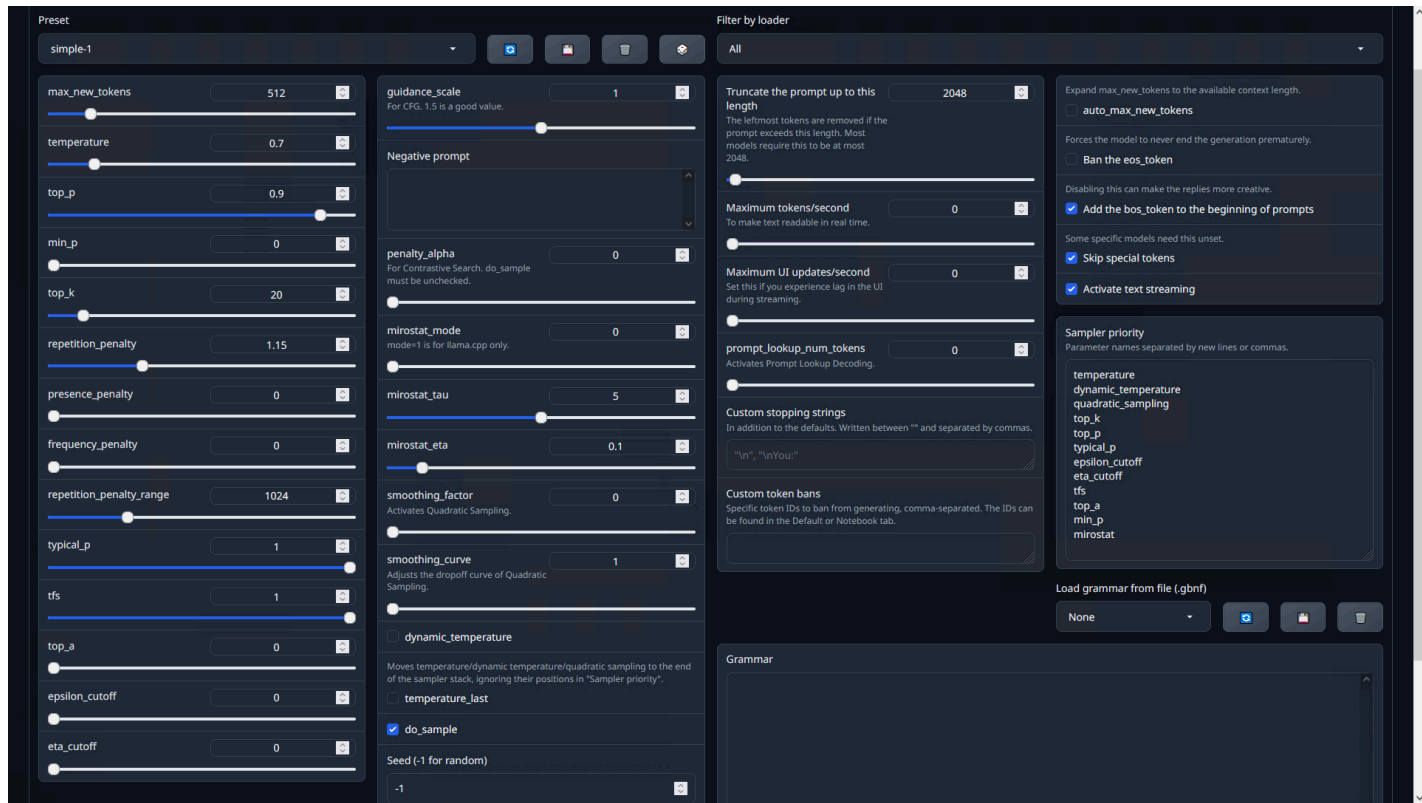


Image 16. Credit: Christopher Fong. A graphical user interface showcasing all inference-time parameters utilized by the LLM.

# Design Summary (with Diagrams)

## Methodology

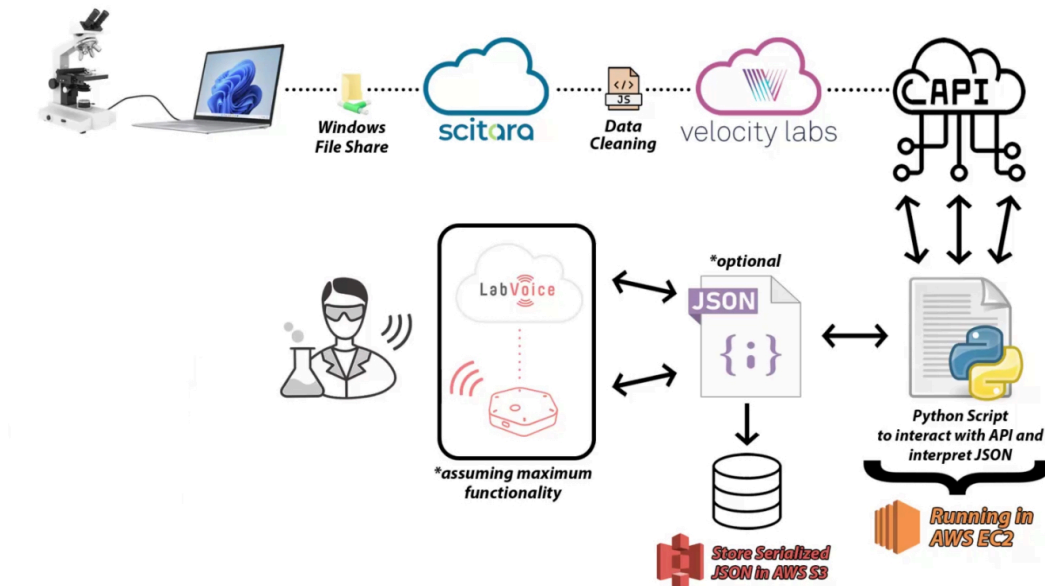


Image 17. Credit: Christopher Fong. Industrial Engineering and Data Science senior design team. A visualization of the MVP methodology.

## Methodology

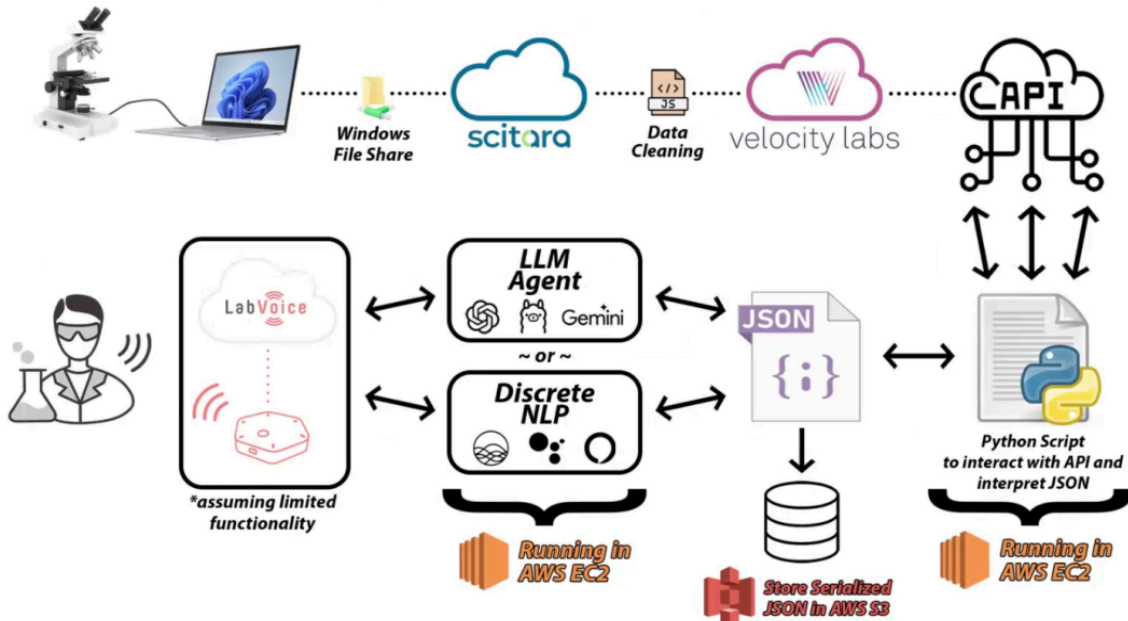


Image 18. Credit: Christopher Fong. Industrial Engineering and Data Science senior design team. A visualization of the methodology beyond the MVP.

## Use Case Diagram

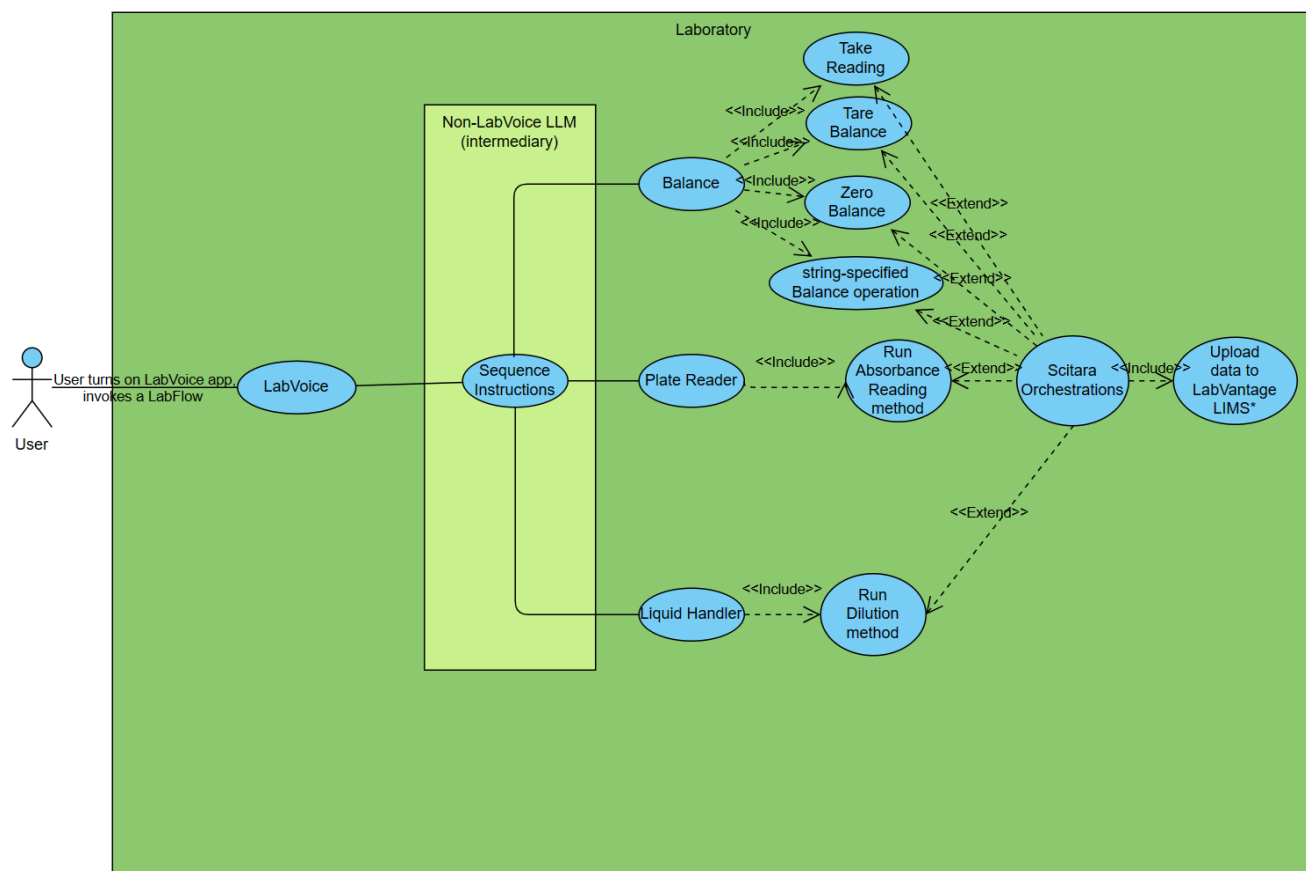
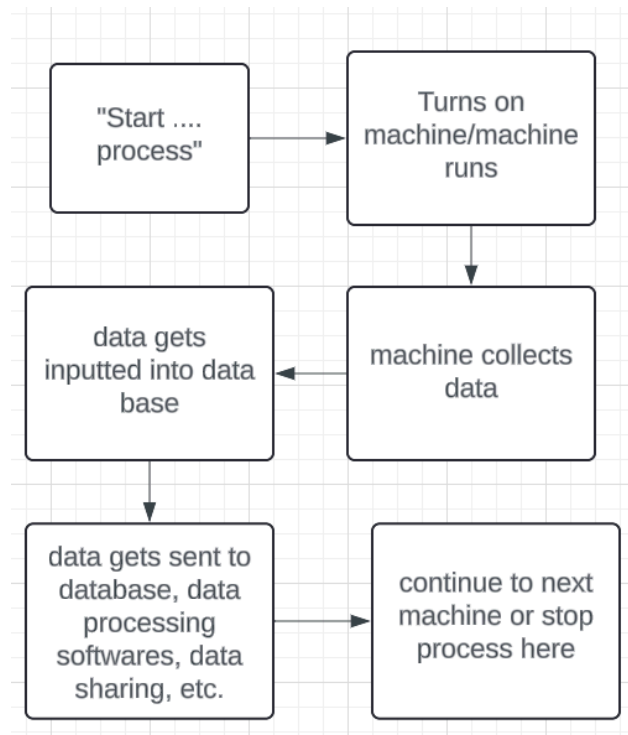


Image 19. A use case diagram demonstrating the finalized product with all laboratory equipment in scope of the overall project.

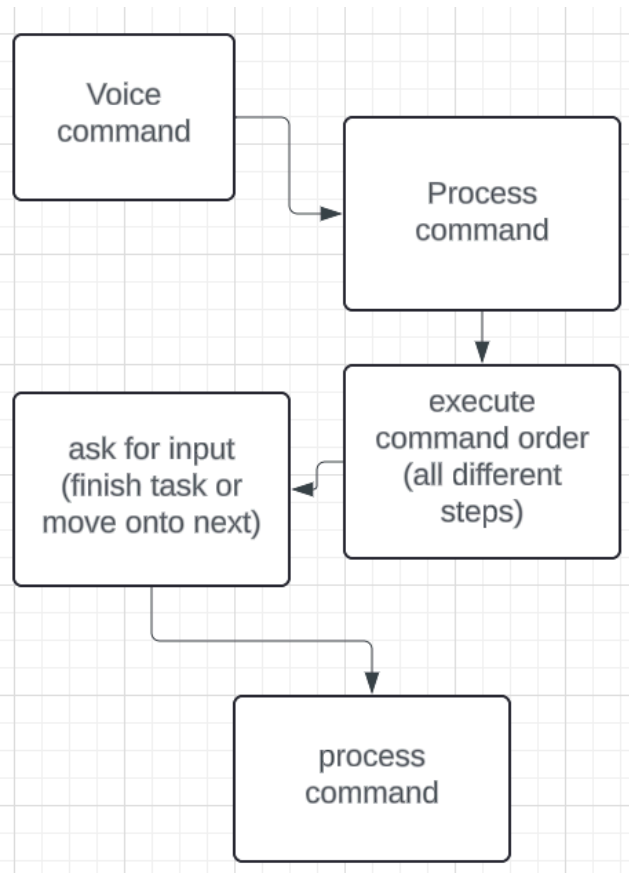
## Activity Diagram



*Image 20. An activity diagram depicting the basic parts of the application process*

The activity diagram serves as a reflection of the Lab Voice lab flows, highlighting the similarity in functionality between the two. Just as the lab flow displays various branches that a node can traverse, the activity diagram follows suit, illustrating the diverse pathways our project activities may take. While the diagram may appear simple at first glance, it encapsulates the fundamental essence of the activities inherent in our project workflow. This visual blueprint provides a roadmap for the basic steps that each user could go through when utilizing our project application.

## Block Diagram



*Image 21. A block diagram. A more simplified version of the activity diagram*

## Design Description

Within the MVP

LabVoice

The MVP hinges on LabVoice's successful integration and implementation on top of the existing workflow the sponsor currently has employed. Utilizing LabVoice's speech-to-text functionality to fully control at least one instrument and report whether the LabFlow run was successful or unsuccessful will meet the requirements of the MVP as expressed by the sponsor.

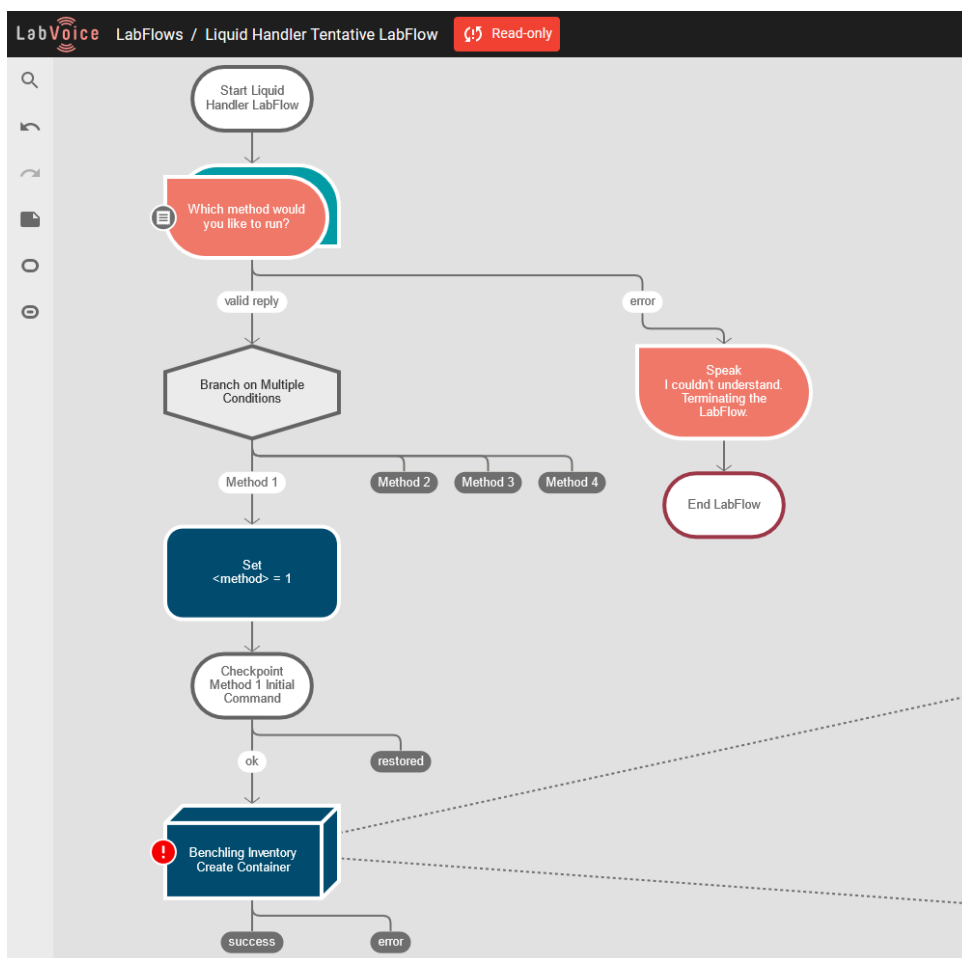


Image 22. Credit: Caleb Gibson. A MVP LabFlow example for the liquid handler.



The necessary requirements for the MVP are likely comparatively simple. In order to develop a LabFlow that meets the sponsor's minimum requirements of initiating an instrument run and an instrument-centered workflow, it will likely require a simple Web service Node with a Scitara execution trigger link supplied as the link to make the HTTP RESTful request to. The Scitara orchestration should be able to handle the remainder of the instrument-centered workflow necessary for the MVP.

Because the sponsor already has a Scitara Orchestration designed to handle the plate reader running, data processing, and cleaned-and-processed data storage as a text file (or potentially a csv, since it appears that the data output by the Scitara Orchestration of the sponsor's is comma-separated, and if not then it can probably be easily made that way, so that it can be displayed by more than just Notepad or any other basic text editor) on the plate reader-connected Windows computer, it should likely be fairly simple to simply call that Orchestration's Scitara trigger execution URL with a LabVoice Node – like the Web service Node or, in case that does not work, a JavaScript "Script" Node that simply visits the URL – should enable complete remote execution of the workflow as desired by the sponsor for the MVP.

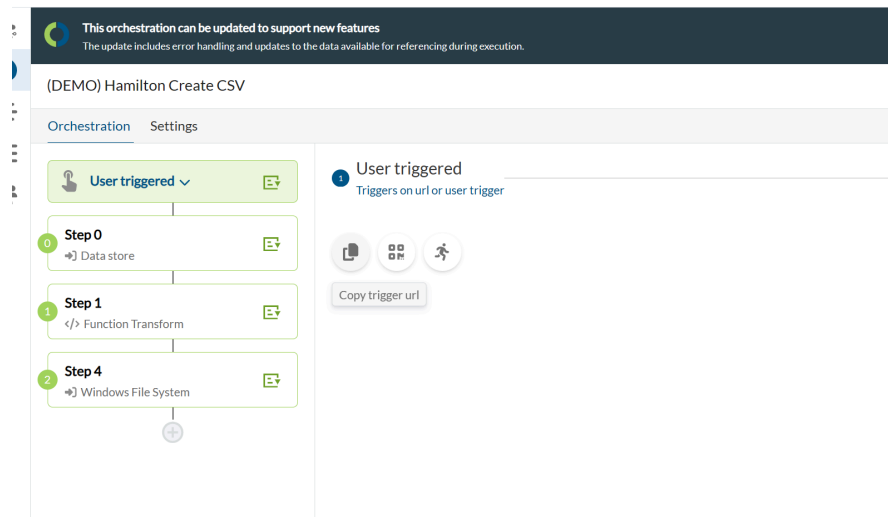


Image 23. Credit: Caleb Gibson. Image illustrating how a Scitara Orchestration can be triggered by a URL.

## ASIS BioTek Plate Reader Test Parser

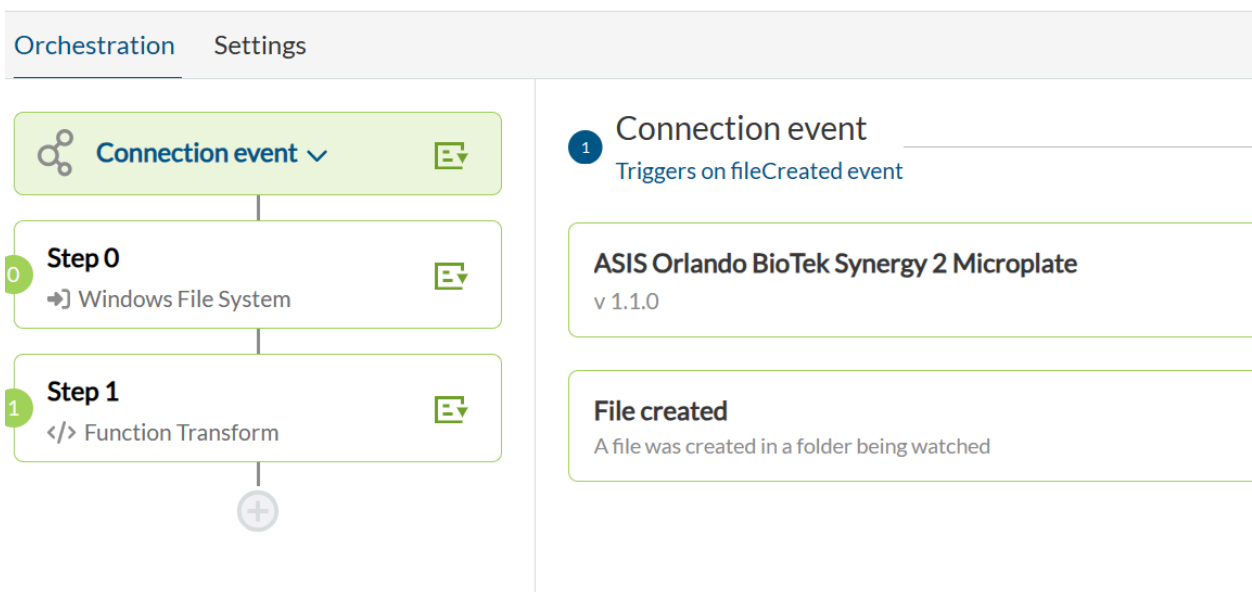


Image 24. Credit: Caleb Gibson. Image illustrating the Connection event for the sponsor's Orchestration that watches the directory on the Windows computer connected to the sponsor's BioTek Synergy 2 plate reader for a "File created" event, and then begins parsing it and cleaning it up to present it in a more human-readable format, and then to put that cleaned-up and processed data in a text file and send that text file to the directory where the initial output data file from the plate reader after a run was completed on it is stored.

## Beyond the MVP

Since the goal expressed by the sponsor is to save as much time as possible from repetitive, manual labor for their scientists, we would like to propose that such a LLM agent be used because of the significant flexibility offered by one, especially when paired with retrieval augmented generation (RAG). RAG allows for new datasets to be accessible to the model without consuming additional memory by storing it into a vector database. This is an extremely quick process that avoids extreme resource allocation which traditional training and fine tuning require.

Having an LLM agent run as a backup to LabVoice's speech-to-text also prevents the headache of the user having to repeat the same thing over and over again when numbers are not interpreted correctly. This is only one example, but there are multiple opportunities where this strategy can be employed to reduce growing pains in the workflow or act as a safety net for other speech-to-text inputs in the proposed workflow example.

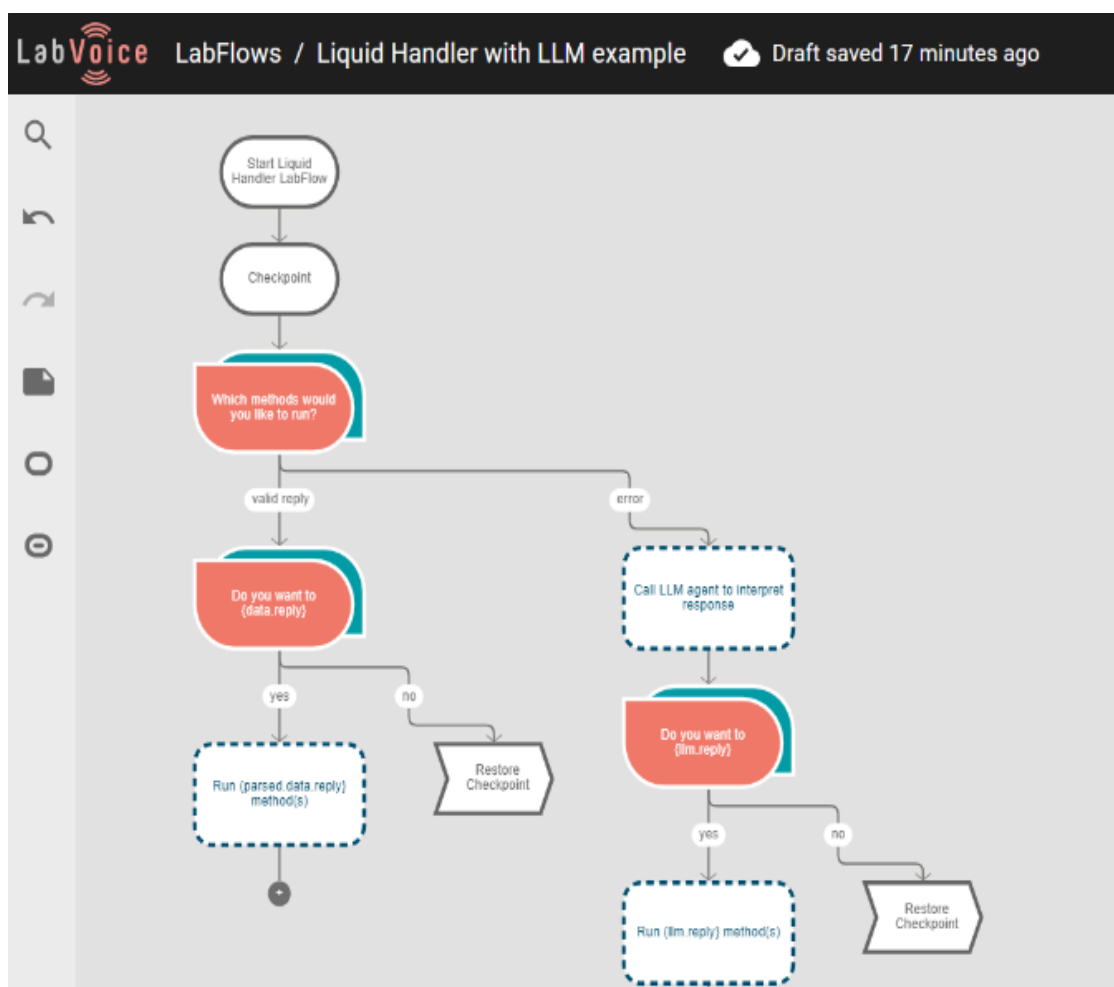


Image 25. Credit: Christopher Fong. A more advanced LabFlow which implements a LLM agent to attempt corrections on speech-to-text numbers captured incorrectly.

Another idea beyond the scope of the MVP that the team could possibly work on implementing is to create LabVoice modes for other languages and regions (in other words, what is known for video games as localization). Considering English is not everyone's first language, users may encounter issues talking to voice-controlled devices if they happen to have a heavy accent or different dialect in the way they speak it. With this may come a limited vocabulary or difficulty in forming complex sentences/commands, resulting in a potential delay of voice recognition systems to transcribe and understand the user's speech. People with speech impairments may face difficulties in using the technology, limiting their

ability to fully use a speech-based interaction device.

While LabVoice supports multiple languages, individuals that come from different linguistic backgrounds with variations in pronunciation or even rhythm may lead to inaccurate interpretations or errors caused in transcription. Difficulties found within operation of the devices that are meant to reduce labor can make users feel uninvolved or left out throughout its usage. This can also cause users to feel frustrated or find difficulties throughout their work process as compared to the typical manual labor approach they may have been used to. Although this is normally a rare occurrence considering these machines are made by taking these potential roadblocks into consideration, the unsmooth process that can be found may lead to mistakes made within the data that is captured. Nonetheless, this can also reduce the overall accuracy and reliability of LabVoice and other voice-controlled devices.

## **Build, Prototype, and Test Plan**

<will eventually include current test results>

### **Build Plan**

As we gear up for the upcoming semester, our team is starting to transition from the research phase to the development phase. With a solid foundation of research under our belts, we are ready to embark on the next leg of this project journey. Our approach to development will start with the pivotal task of setting up API endpoints. This serves as a crucial step in facilitating seamless data transfer between machines and the database.

Over the past few weeks, we have been actively engaged in communication with the project sponsor, delving into the specifics of API endpoint-related matters. This collaborative effort aims to establish a robust framework for data exchange, ensuring that our development efforts align closely with project objectives and sponsor requirements.

Simultaneously, we plan on building up the groundwork for incorporating Lab Voice into our project framework. We currently have lab flows based on the example lab flows provided by the Lab Voice team, but we have yet to create a lab flow that is more specific to our project. So this would involve creating our own independent Lav Voice Lab flow. By building these flows, we aim to efficiently integrate various software components, enhancing the functionality and usability of the ASIS Accenture Lab Assistant platform.

Our build plan takes a lot of inspiration from the diagrams showcased in the design summary, serving as a blueprint for our development efforts. By adhering closely to these design specifications, we strive to tailor our solution to meet the unique requirements and preferences of the project sponsor. As we start this development phase, our team remains committed to delivering a robust and innovative solution that not only meets but exceeds the expectations of our stakeholders.

## Prototype Plan

When considering our prototype plan, the scope of our task differs slightly from the usual projects and their prototyping endeavors. Given that we already possess an application capable of managing the frontend aspects, our focus shifts towards refining and augmenting its functionality to align with project objectives. While there isn't a conventional prototype to construct per se, our aim is to develop a robust Lab Voice application near its final iteration that seamlessly integrates with the laboratory instruments.

This Lab Voice application, though in a rough draft stage, will help significantly progress us towards our ultimate goal. By ensuring its compatibility with laboratory instruments and optimizing its performance, we lay the groundwork for extensive testing and validation in subsequent phases.

Although the prototype is unconventional in nature, it serves as a useful tool for implementing our testing plan. Its functionality will be rigorously assessed, allowing us to identify any shortcomings or areas for improvement before proceeding further. Through iterative refinement and validation, we aim to fine-tune the Lab Voice application, ensuring its efficacy and reliability in real-world laboratory scenarios.

All in all, our prototype plan revolves around refining and enhancing existing capabilities, culminating in the development of a Lab Voice application that forms the cornerstone of our testing and validation efforts, Through this iterative process, we inch closer towards realizing the vision of the ASIS Accenture Lab Assistant project which aims to revolutionize laboratory operations with its innovative functionality.

## Testing Plan

In outlining our testing plan, we recognize the importance of leveraging the laboratory environment at our disposal to the most that we can to ensure the efficacy and reliability of our application. Our strategy revolves around soliciting feedback from key stakeholders within the Accenture ecosystem, encompassing lab researchers and personnel across various teams.

Initially, testing will be conducted internally, involving our own team, the IE/DS team, and the accessible project sponsor. This internal testing phase serves as a crucial step in validating the functionality of our application, specifically with its responsiveness to voice commands which is a core aspect of our project. By eliciting feedback from individuals directly involved in laboratory operations, we aim to identify any potential struggle points or areas for improvement early on in the development process.

Moving forward from that stage, we will extend our testing efforts to include the external partner company collaborating with Accenture on this project. This phase mirrors the internal testing process, with the objective of garnering insights and feedback from an external perspective. By involving



the partner company in our testing regimen, we ensure that our application aligns with their expectations and requirements, further enhancing its suitability for real-world deployment.

Furthermore, we acknowledge the importance of ensuring that our application meets the standards and preferences of both sponsor companies involved in the project. By soliciting feedback and approval from these entities before finalizing the project, we aim to deliver a product of highest quality. A product that not only meets but exceeds the expectations of our stakeholders.

Looking ahead, we are aware that the sponsor's have an ultimate goal of showcasing this product at a convention. Therefore, we understand that it is imperative that our application is meticulously tested and refined to ensure that it is in optimal condition for presentation and deployment.

Essentially, our testing plan aims to go for a thorough strategy that confirms the effectiveness, user-friendliness, and appropriateness of our application for a wide range of stakeholders. By conducting iterative testing and refining our approach, our goal is to create a product that fulfills the requirements of our sponsors but also aids in the innovation and excellence in laboratory automation.

As a disclaimer, it's important to note that all the outlined plans are currently in a tentative state and are subject to adjustments as needed. However, we intend to adhere to these plans as a foundational framework for our project execution. Among these, the testing plan holds particular significance, as it represents a robust and optimal approach to validating the functionality of our project.

While remaining flexible to potential changes, we recognize the value of maintaining stability in our testing strategy, given its pivotal role in ensuring the quality and effectiveness of our application. Therefore, the testing plan will serve as a cornerstone, providing a reliable roadmap for our testing efforts through the project lifecycle.

The build plan is slated for further refinement in the lead up to the summer semester, with the goal of achieving a more comprehensive and detailed framework before embarking on the development activities. This deliberate approach seeks to ensure a seamless transition into the development phase, laying the groundwork for a productive and efficient execution of the project objectives. While acknowledging the provisional nature of our plans, we are committed to leveraging them as a basis for our project execution.

# **Conclusions**

## **Characterization of Results**

<we will add results once we have begun development in the summer semester>

## **Summary of Project**

As reiterated, the ASIS Accenture Lab Assistant project is an ambitious endeavor centered around harnessing the power of AI interface, LabVoice, in conjunction with Accenture's extensive data resources. The overarching goal is to construct a cohesive product that empowers researchers to streamline their laboratory operations effortlessly. With a mere vocal command, researchers can kickstart processes, activate instruments, gather data, facilitate data transfer, and initiate data analysis seamlessly.

This transformative approach not only simplifies lab procedures, but also liberates researchers from mundane tasks such as turning on a machine and getting it set up. This affords them invaluable time to delve into deeper explorations. By automating routine tasks, the Lab Assistant accelerates the research trajectory, propelling breakthroughs and discoveries at an unprecedented pace.

In conclusion, the ASIS Accenture Lab Assistant epitomizes efficiency and innovation, revolutionizing the research landscape by infusing it with unprecedented accessibility and agility. Through the fusion of AI technology and Accenture's wealth of data, it heralds a new era of research excellence, where every voice command unlocks boundless potential and propels scientific inquiry towards uncharted horizons.

# **Administration**

## **Budget and Financing**

When it comes to managing the project's budget and financing, Accenture's innovation lab operates uniquely compared to other innovative companies out there in the field. Accenture's innovation lab does it in a different manner that fosters collaborative partnerships with other companies to co-create cutting-edge innovations and technologies. Since the environment is so dynamic, and Accenture is teamed up with other investors, the ASIS Accenture Lab Assistant project is afforded a significant advantage: access to a suite of essential software programs including LabVoice, Scitara, and other instrumental software tools. The most advantageous part, is that this access comes at no cost to the company or the project team, facilitated and scalable through Accenture's robust internal operations. Not only does this streamline our workflow, but it also ensures that we can leverage the latest technologies without the financial burden usually associated with such resources.

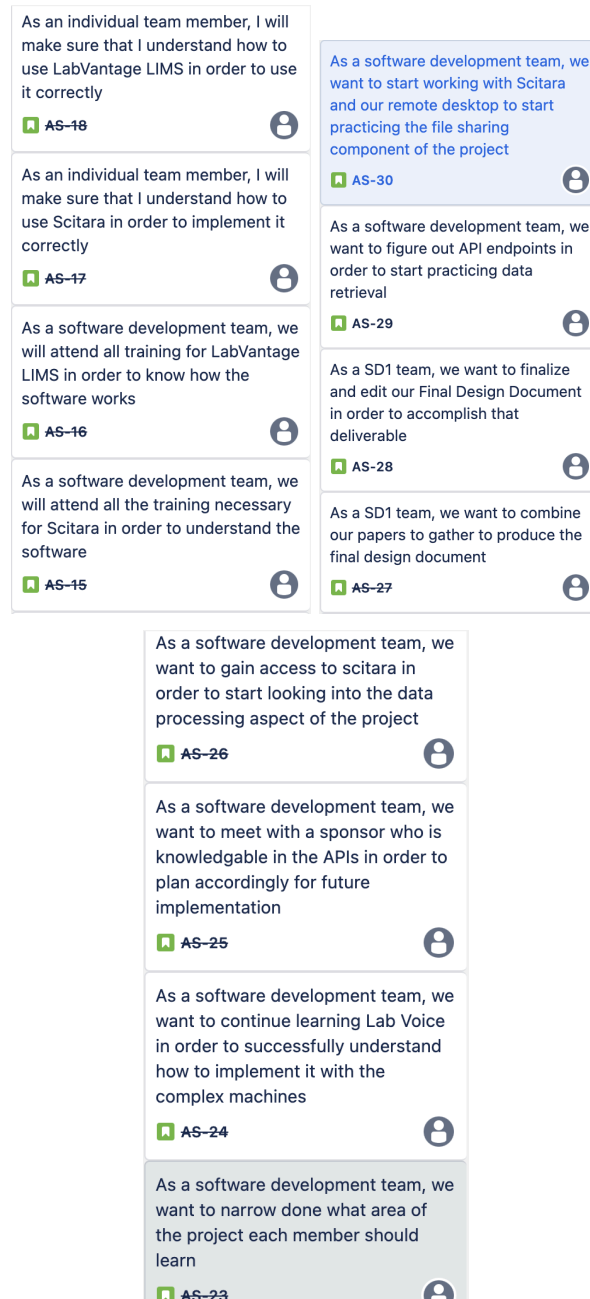
As of right now, our team lacks precise information regarding the budget and financing specifics. Nevertheless, we anticipate the ability to acquire these additional insights in the future through discussions with our main sponsor's colleagues, potentially shedding light on partnership agreements and related intricacies. Currently, our team lacks concrete statistics or figures. However we can gather and share pertinent details as they become available, to ensure transparency and informed decision-making as we progress with our project.

# Backlogs

## Product Backlog



Image 26. Credit: Edrienne Gregana. First part of the team's overall issues throughout the whole spring semester



*Image 27. Credit: Edrienne Gregana. Second part of the team's overall issues throughout the whole spring semester*

## Current Sprint Backlog

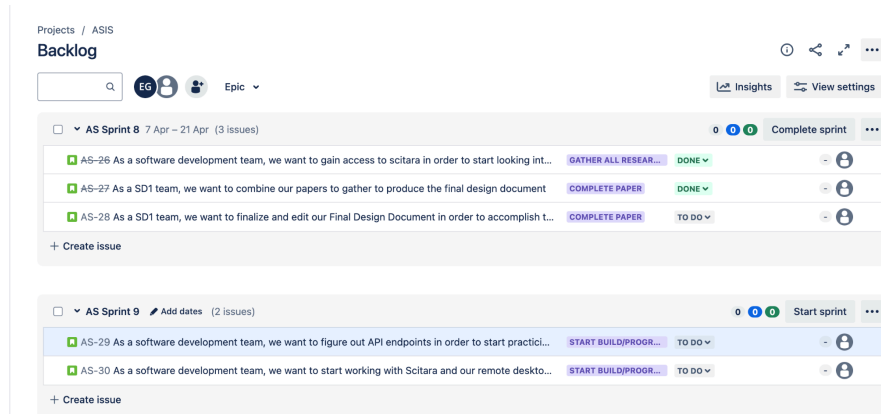


Image 28. Credit: Edrienne Gregana. The team's current sprint backlog that we are working on completing

## Milestone Chart

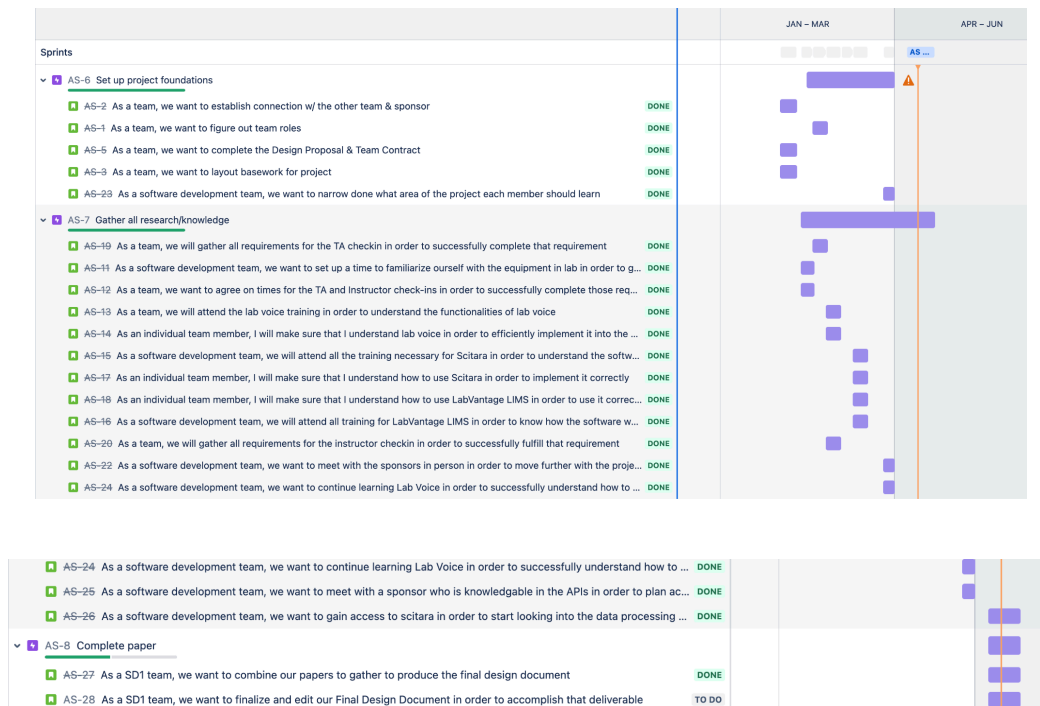


Image 29. Credit: Edrienne Gregana. The team's Jira timeline of everything we've been working on for the entirety of the Senior Design I term.



## Schedule

| Issue                                                                                                                                                                                                                                                     | Date/Timeline        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Get all administrative tasks done<br>(set team roles, set up Jira,<br>establish connection with sponsor,<br>write up team contract, etc.)                                                                                                                 | February 5-12        |
| Get groundwork done <ul style="list-style-type: none"><li>- View the lab in person</li><li>- Get familiarized with<br/>technologies and softwares</li></ul>                                                                                               | February 12-18       |
| Attend Lab Voice training and<br>understand the basics of the<br>software                                                                                                                                                                                 | February 18-March 4  |
| Practice using Lab Voice and create<br>sample lab flows                                                                                                                                                                                                   | March 4-March 10     |
| <ul style="list-style-type: none"><li>- Attend Scitara informational<br/>meetings and get necessary<br/>information</li><li>- Attend informational<br/>training/meetings for<br/>LabVantage LIMS in order to<br/>understand its functionalities</li></ul> | February 19-March 26 |
| Figure out information regarding<br>APIs in order to start working<br>through data processing                                                                                                                                                             | March 26-Apr 7       |
| Complete our Final Design<br>Document for the first semester                                                                                                                                                                                              | Apr 7-Current        |
| Start Development                                                                                                                                                                                                                                         | Next semester        |

The table provided outlines our tentative schedule for the semester of Senior Design I, delineating our proposed roadmap for project execution. Initially, our efforts were concentrated on gaining a foundational understanding of the project landscape, including familiarizing ourselves with the technologies and machinery prevalent in the laboratory setting. With this basic understanding, we proceeded to delve deeper into our research, employing methodologies elucidated in our research section to unearth insights and refine our project approach.

Looking ahead, our focus will shift towards development. With a pre-existing UI environment at our disposal, the emphasis shifts away from intensive app development, affording us the opportunity to channel our efforts towards the data processing facet of the project. While it is not a mandatory component of our project scope, we harbor an interest in integrating generative AI. This is a concept that piques our interest and holds potential for enhancing the functionality of this project and its outcomes. While the specifics of carrying this out are still up in the air, it is a direction that we aspire to explore further in the following semester if time permits.

## Setbacks

There are various setbacks that the team has encountered during its progress towards the mission. A prime example is coordination between the various teams working on this project, which include not only this team and the sponsor's team, which seems to have more and more members as time goes on, but also the Industrial Engineering and Data Science senior design team that was tasked by the sponsor with contriving a workflow to be optimized by the voice assistant and specifying how it could specifically be optimized by it, as well as LabVoice development and integration support teams. Coordinating efforts and communications and arranging meetings between the CS SD team, the IE/DS SD team, Accenture, and more recently with the LabVoice development and support teams has proven to be quite difficult due to extensive schedule conflicts between the members of each team.

Moreover, due to a lack of access to the instrument software and information about it, conflicting schedules, the busyness of the sponsor's technical members, lack of remote desktop access until just prior to Spring Break, which required traveling to the sponsor's lab in-person to configure it (this required some schedule rearrangement and frontloading of work hours in the week), lack of Scitara access until Friday, April 12, 2024 (due to either security concerns or simply busyness on the part of the member of the sponsor's team who could grant access to it, Shabbir), and other factors, it has been difficult to conduct research and true development. Much of the software that the sponsor desires us to employ and develop with has frequently no or very rarely very few accessible online resources, as much of it is proprietary, and the companies that own said software do not provide documentation on their websites or another accessible manner that enables

independent research and development. The UCF CS SD team has had to rely on the sponsor to obtain meetings with the providers of the various software products for it. So far, a connection with the LabVoice team for the UCF CS SD team was secured and, during Spring Break, a light connection with the Hamilton vendor. Thankfully, these facilitate integration and development, but regardless, many of the fine details that the team needs to know – for instance, the keys in JSON objects returned by measurements or methods of scientific instruments in the lab – remain elusive or extremely difficult to uncover.

Doing physical test runs with the Accenture team's instruments is challenging, as if an instrument encounters an error mid-run, even if it was started remotely, it may demand human intervention in-person to mend, depending on the error that occurs. Being physically present in the Accenture team's lab is something that their business hours and the UCF CS SD team members' schedules have rarely allowed. In spite of this, some progress has been made.

Regarding hindrances more specifically that obstruct or slow down instrument-LabVoice integration, after conducting some research on LabVoice and its capabilities, it was discovered that its ability to integrate directly with instruments is highly contingent upon whether those instruments have APIs or not. On the bright side, the two main instruments the sponsor prefers that the team work on both have software applications – BioTek Gen5 PC for the plate reader and Hamilton VENUS for the liquid handler – and are connected to Windows laptops at all times to host their software. Unfortunately, neither of them appears to have software with an accessible API, let alone a RESTful API. The Accenture team is working with the Hamilton vendor to upgrade the software for their Hamilton Microlab

Nimbus liquid handler to the latest version, which has an API. This progress was notified to me by the UCF CS SD team's main point of contact on the Accenture team, Mr. Michael Kelliher, on March 14, 2024. According to someone on the LabVoice development team, Mr. Tobias Rothe, "Those instruments [that don't have APIs] are terrible to work with."

LabVoice has 185 built-in Nodes available to Accenture, but very few (fewer than 10) are designed for working directly with instruments so well that they abstract away the API call portion of it. The reason for this is the previously mentioned diversity in instruments and in the software that accompanies them.

This is why when a new avenue to explore instrument integration was opened, one with more versatility and far more flexibility and ease of use, at least on the surface, during a meeting on March 29, 2024, it was immediately desired to be pursued by the team and by the member of the sponsor who proposed utilizing it, Mr. Chris Angelelli. Mr. Angelelli had been unable to meet with the UCF CS SD team for several weeks due to business-affiliated timing conflicts and personal matters. This new avenue, indirect instrument integration by relying on Scitara's DLX platform, was seemingly optimal. However, it has yet to be fully explored, as access to it was only granted to the UCF CS SD team – and, even then, only one member of it, so that this member had to create login credentials and share them with his fellow group members – at approximately 5:00 P.M. Standard Eastern U.S. time on Friday, April 12, 2024. As such, very little time has been afforded to the team to create Orchestrations on it, test calling them with LabFlows, and perform runtime evaluations of those Orchestrations with LabVoice, particularly as this document was due approximately a mere two days after this access was granted. Regardless, in spite of all this, it remains to be seen whether indirect instrument integration by having LabFlows call

Orchestration URLs is an advantageous and optimal integration, as opposed to using direct LabFlow integration with the instruments in the lab.

# **Acknowledgments**

## **Assistance from the Sponsor and Others**

### Sponsor

The sponsor has provided extensive support throughout the entire duration of the project. The sponsor has explained to the team much of the intricate workings of their various laboratory systems and subsystems, which the team was previously almost entirely unfamiliar with for the most part. The sponsor has explained the various scientific processes they go through and what they are endeavoring to optimize. They have demonstrated several of these processes in-person to the team, using the unique equipment in their lab. Some more technically oriented members of the sponsor's team, like Mr. Shabbir Kheraluwala, have also demonstrated virtually some of the sponsor's existing semi-automated workflows, and presented in-depth diagrams explicating how those workflows progress, with labels for each step of the process. Interestingly, the workflows shown include but are not limited to semi-automated workflows for the sponsor's Hamilton Microlab NIMBUS liquid handler, which is the most complex of the instruments that the sponsor has asked the team to develop the voice assistant integration for.

These workflows are not planned on being directly used by the team for the integration of the voice assistant with the instrument-centered workflows in the sponsor's laboratory. The rationale for this is that another technically oriented member of the sponsor's team, Mr. Robert Ellis, has thus far indicated that the task these workflows accomplish, while satisfactory as a proof of concept for the notion of remotely starting and completing an instrument and storing the output data on the Internet, does not meet their

clients' stronger urges that the remotely commanded instrument-centered workflow conclude with data storage in the laboratory's LIMS or ELN system. Nevertheless, these workflows are far from useless. They provide an excellent springboard from which to develop other LabFlow-instrument remote execution and error handling integrations.

The sponsor has also generously provided members of the team with access to a multitude of proprietary tools. Among these tools are LabVoice and Scitara. LabVoice is the platform on which the voice assistant is to be developed and on which it executes. The core notion of the project – voice-commanded instrument-centered workflow automation and post-instrument-operation-execution Internet data storage and retrieval – hinges on it. Scitara is a fantastic centralized automation tool that the team leverages to run the instruments indirectly. With Scitara, the direct interaction with instrument software – or, in the case of serial-communications-based devices like the sponsor's Mettler-Toledo digital balance – is somewhat abstracted away and is handled by Scitara. In exchange, Scitara handles many aspects sought by the sponsor, like audit trails and sample tracking, automatically. That way, what is left is manipulation of JSON objects that represent vast quantities of data present in even the simplest of workflows. On top of that manipulation also is high-level connection flow, and associated security, authentication, and even authorization, logic. That manipulation and complementary connection flow and security, authentication, and authorization logic is handled by the user, and the sponsor has demonstrated this and several other aspects of Scitara via multiple hours of instruction and both in-person and virtual tutorials, largely provided by Mr. Angelelli.



Regarding LabVantage, the sponsor's bundled LIMS and ELN system provider, the sponsor has also provided assistance to the team in comprehending, at a very high level, what interactions are necessary for a product beyond the MVP. In particular, Mr. Ellis has been very helpful in explaining the primary connection point categories that the team will need to interact with in order to successfully fully orchestrate and integrate an instrument-based workflow with LabVoice and LabVantage such that at instrument completion, the data output from the run by the instrument gets automatically uploaded and sent to LabVantage for remote data access, storage, and retrieval. Such remote electronic archival and recovery is essential for Accenture's clients, who are somewhat often – as mentioned previously – working in industries under heavy regulatory scrutiny, and are prone to face audits encompassing all aspects of their work, from research to production to quality control and testing.

## LabVoice Team

The LabVoice team's support and knowledge is instrumental in comprehending LabVoice in many respects. For example, Mr. Tobias Rothe of LabVoice, a developer on their team, has uploaded examples of using proprietary instruments for direct LabFlow-instrument integration from other organizations, like Unchained Labs. The LabVoice team also gave a one-and-a-half-hour-long tutorial to the team that a couple of members of the Accenture team, including Mr. Kelliher and Mr. Ellis, participated in as well. Mr. Rothe has answered a plethora of questions that members of the team have had. For instance, during one team member's investigation of the aforementioned LabFlow that he had provided to illuminate what using a LabFlow to directly control an instrument might look like, the team member came across a method of making an API call that they were unfamiliar with,

and whose configuration varied from any HTTP request API (like Fetch or Axios) that they had seen. They asked Mr. Rothe about it, and he informed them that the HTTP requests API that this LabFlow was using was called GRPC<sup>8</sup>. He explained that it was similar to any other HTTP requests API, but that it was likely harder to work with because it does not employ text-based payloads like JSON but instead relies on pure binary. He explained that its advantage is that it makes the API calling and receiver-end handling faster, since pure binary code is used, but also more difficult to work with from a developer perspective. Consequently, it was able to be decided that the time could try to refrain from using it, since it is still similar to other HTTP request APIs in terms of functionality, but more difficult to work with and find resources for online which is a disadvantage that outweighs its slight speed advantage.

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<sup>8</sup> Google Remote Procedure Call

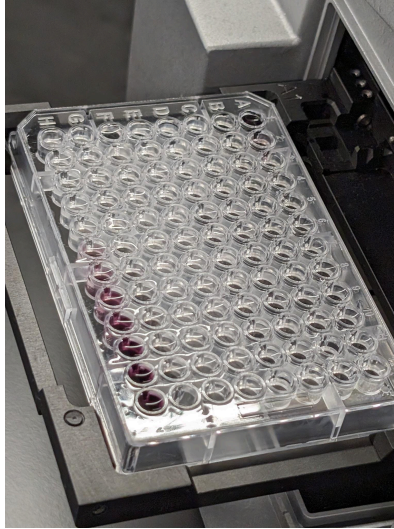
## Unique Facilities and Equipment

### Accenture ASIS Orlando Laboratory

The sponsor owns a laboratory on the fourth floor of a building located at the address 9400 Southpark Center Loop, Suite 400, Orlando, FL, 32819. This laboratory is what had been called by Mr. Kelliher as a “showcase” or an “innovation” lab. Here, they can mimic all the same processes that many major pharmaceutical manufacturers and other clients use to develop their products, at least for most of the processes that these industries and major pharmaceutical manufacturers would perform. However, to avoid having to deal with the administrative headache that red tape compliance is, the sponsor pursues a far more benign and safe method of carrying out these common highly-sensitive and highly-potentially-dangerous processes — beer brewing and distillation, and using food coloring to do experiments. For example, when doing a plate reading using the BioTek Synergy 2® plate reader to obtain a UV-Vis<sup>9</sup> absorbance measurement of all the wells in a 96-well plate, the sponsor utilizes food coloring, rather than a bioluminescent sample that one of their clients might utilize. (Subsection continues on next page)

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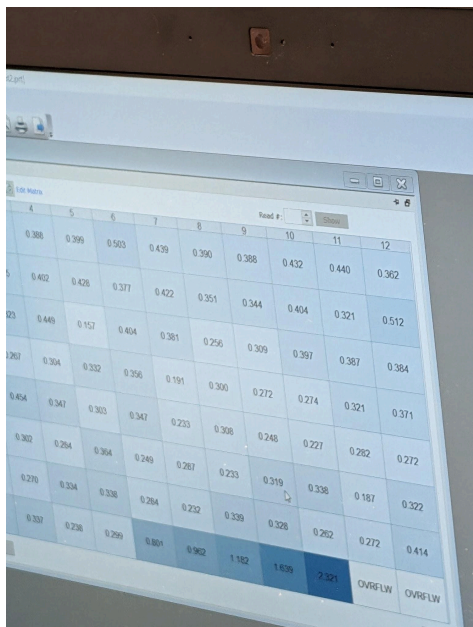
<sup>9</sup> Ultraviolet-visible. Ultraviolet-visible absorbance spectroscopy has to do with the light in the ultraviolet and near-ultraviolet visible light spectrum absorbed by a substance.



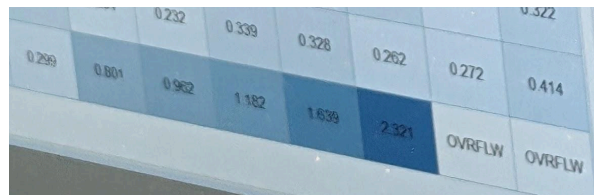
*Image 30. Credit: Caleb Gibson. A 96-well plate, loaded on the BioTek Synergy 2® plate reader's plate rack. On the top right (viewed from this perspective) of the plate, one can observe a very highly saturated well of "Accenture Purple" food coloring. Similarly, on the bottom left (viewed also from this perspective) of the plate, one can observe a series of seven wells filled with varying concentrations of Accenture Purple food coloring. These wells visibly exhibit increasing concentration, which is correlated with absorbance, at least in the case of food coloring, descending.*

The laboratory has a plethora of instruments and equipment. The main ones of relevance to the UCF CS SD team are the sponsor's BioTek Synergy 2® plate reader, Hamilton Microlab NIMBUS automated liquid handler, and Mettler-Toledo digital balance. With regards to the Mettler-Toledo digital balance, that device relies on serial communications for communicating with Internet-based applications. So, to allow serial communications between the Scitara DLX platform and the balance, an IoT gateway box, provided by Scitara, is physically connected to it.

## BioTek Synergy 2 Plate Reader



*Image 31. Credit: Caleb Gibson An image depicting the table of results from taking UV-Vis absorbance measurements using the BioTek Synergy 2 plate reader with the 96-plate well shown in Image xx. This table is produced as part of the GUI associated with the BioTek Synergy 2 plate reader's companion Windows application, the BioTek Gen5 PC software app.*



*Image 32. A more close-up view of Image 31's bottom right half. This section of the output table corresponds to the bottom left series of wells of varying Accenture Purple food coloring concentrations, whose concentrations increase descending in Image 30, or from left to right here. Notice how the absorbance readings for the two most concentrated wells, which occupy row H, columns 11 and 12, do not contain a numerical value but contain a string value "OVERFLOW". The "OVERFLOW" reading indicates that the well was filled with too much liquid, and so an accurate absorbance measurement could not be taken.*

As its name suggests, the BioTek Synergy Plate Reader is a versatile Multi-Mode Microplate Reader used for both research and development and in vitro diagnostic applications. Equipped with a range of tools including a combination of monochromator, filters and dichroic mirrors, the Synergy 2 offers a solution for many analytical tasks. One of the noteworthy features of

the Synergy 2 is the three broad-spectrum light sources which have been chosen for optimal illumination in all applications.

To recap, the Accenture team has two main instruments that they want the CS SD team to work with: the plate reader and the automated liquid handler. The plate reader in the sponsor's lab is a BioTek Synergy 2 plate reader. This is a somewhat older instrument, but is nevertheless useful and comes with a desktop application powered by BioTek's Gen5 PC software that can be accessed and utilized by connecting the plate reader to a Windows machine, which the sponsor has accomplished by connecting it via a wired connection to a laptop computer running Windows. The plate reader is considered by the sponsor to be the simpler of the two instruments and probably the easier of the two main, more sophisticated instruments to integrate LabVoice successfully with.



*Image 33. Credit: Christopher Fong. The sponsor's BioTek Synergy 2® plate reader.*

So far, details regarding the plate reader are scarce, but some documentation exists for it on the Internet, which at least one member of the team has read some of. Most notably, the team member has read documentation on its software's generated error codes, which is likely the main relevant component of the available online documentation for it for the team's purposes.

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 050x | <p><b>Light beam saturated (too much light). Relative Fluorescing Units (RFU) reached (99999).</b></p> <p>This error can indicate one of several scenarios. It is possibly due to incorrect chemistry, e.g., the fluorescence standards dispensed to the plate exceed expectations.</p> <p>See also the description for 0E01.</p> <p>Verify that the microplate carrier access door is closing properly and no ambient light is getting into the read chamber.</p> <p>If you are manually setting the Sensitivity values in your Gen5 procedure, try using the Automatic Sensitivity Adjustment feature.</p> <p>Verify that the Gen5 Fluorescence/Luminescence filter table matches the contents of the filter wheels. Verify that the EX filter in position #2 does not overlap with the EM filter in position #3.</p> <p>For models with injectors, the internal chamber may require cleaning (see <b>Preventive Maintenance</b>).</p> |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

*Image 34. A sample error code string and description from some of the limited BioTek Synergy 2 plate reader documentation available online.*

Based on meetings this team member has independently completed with the sponsor and the LabVoice team, specifically with Mr. Kelliher and Mr. Rothe, in conjunction with this documentation, they have determined a likely solution to error handling for the plate reader. When the team has a more extended period of access to the plate reader and its software, a plan has been proposed to program the plate reader's software to, if the plate reader emits an error in the midst of a run that was commanded by LabVoice, send an email to the email address of the user who invoked the LabVoice workflow – or "LabFlow," as it is technically called – that commanded it to run, with the email containing the error code associated with the error that the plate reader encountered.

In terms of a usual run with the plate reader's Gen5 software, the main parts that are relevant are the serial number of the device connected to, its status, the temperature (somewhat), then the actual starting of the read operation and the returned data. A UV-Vis absorbance reading with a 96-well plate returns a matrix with floating point values in each cell, or, in the event that too much liquid was present in any of the wells for the



reading, a string with the value, "OVERFLW" (used because the well has overflowed with the liquid sample in question).

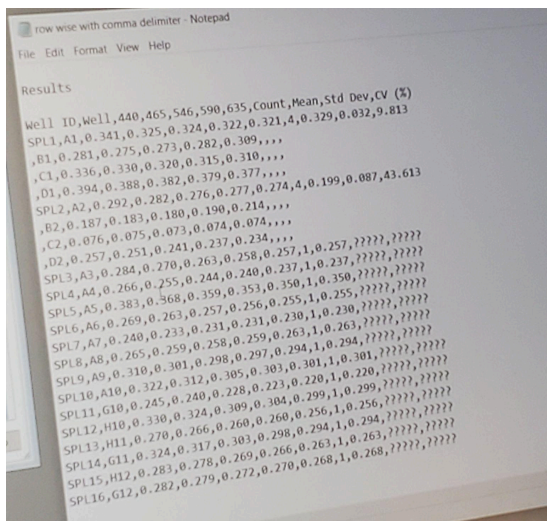


Image 35. A text file containing output and Scitara-processed data for a 96-well plate after running an absorbance reading operation with the plate reader.

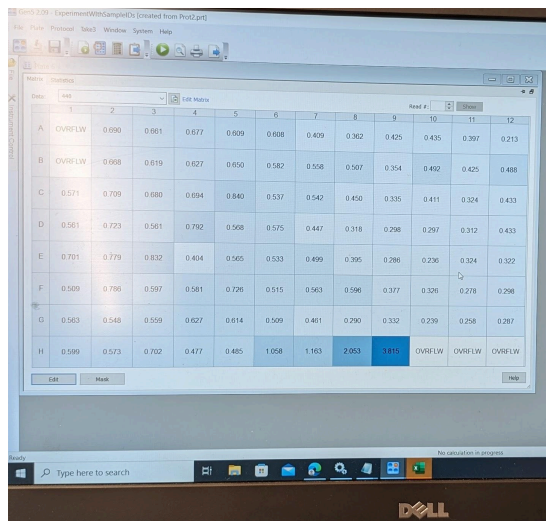


Image 36. An image of the output matrix in the Gen5 PC software after a different absorbance reading run.

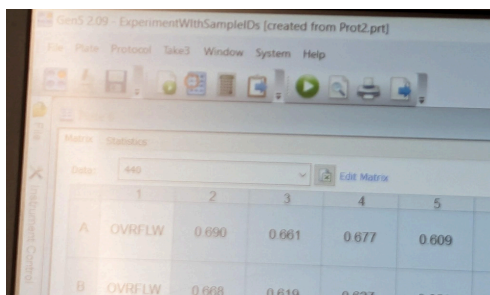


Image 37. A closer-up view of the columns in the matrix and their values. "OVERFLW" means the sample well had too much liquid. Notice that the procedure type file uses the extension, "\*.prt".

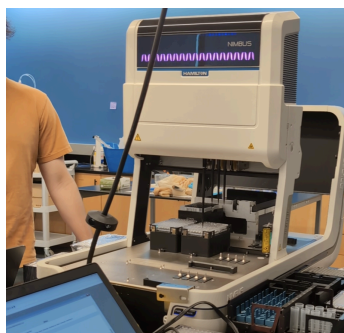


## Hamilton Microlab NIMBUS Automated Liquid Handler

The automated liquid handler, on the contrary, is a Hamilton Microlab NIMBUS automated liquid handler. It is a far more complex machine, with a more involved and complicated software application, called VENUS, accompanying it. At least one team member has read some of the VENUS Programmer's Manual, an over-400-page piece of documentation provided to two members of the UCF CS SD team by Mr. Mitchell See, another member of the Accenture team, on March 20, 2024. The most useful details to be extracted thus far have been some of the utility methods, such as the "SendEMail" method, and error-handling configuration capabilities, as well as the potential for third-party libraries to be imported and linked for methods that invoke them. These elements are discussed in further detail later.

Whereas the BioTek Synergy 2 plate reader's software is simplistic by comparison and does not provide nearly as much room for customization in running procedures, the data output, and the like, the Hamilton liquid handler possesses a software package that enables scientists to develop and invoke custom methods that the liquid handler can run. These methods are contingent upon a broad range of parameters, including but not limited to plate count, well count on each plate, plate placement and sequencing on the liquid handler track, pipette tip rack starting row, liquid sample volume per well or well group, which plate and wells to extract liquid from and which plate and wells to move liquids too, which measurements, if any, to perform when pipetting liquid samples across plates – for instance, measuring the surface tension of the liquid being pipetted, and which directory to store output data into and even how to store it (the default is apparently an Excel sheet, or a .xlsx file, although this can be customized).

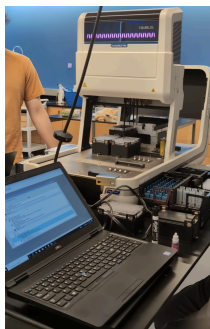
However, many of the methods developed and used by the Accenture lab team for the automated liquid handler that members of the team have seen so far during some of their visits to the laboratory and independent research on the sponsor's configuration of the VENUS software do not require extensive parameterization, and may simply be run employing a simple sequence of preset options. Although members of the team have only witnessed relatively simple liquid handling processes, like the "Dilution" method, this could facilitate direct instrument LabFlow integration. However, it has been elucidated by Mr. Chris Angelelli that allowing remote automated control of the Hamilton Microlab NIMBUS automated liquid handler would likely be far more difficult than merely making a REST API call, should the sponsor successfully upgrade their VENUS software to the latest version<sup>10</sup>, dimming hopes of direct LabFlow integration with the Hamilton Microlab NIMBUS automated liquid handler, if any integration is to occur at all. Regardless of the improbability of this prospect, it was explored further prior to Mr. Angelelli's input, especially using the VENUS Programmer's Manual provided by Mr. See.



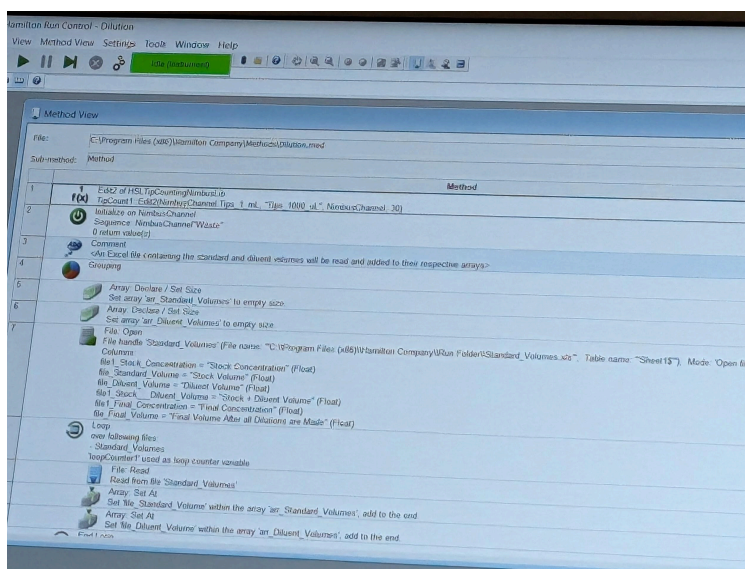
*Image 38. Credit: Caleb Gibson. The Hamilton Microlab Nimbus automated liquid handler during a run of the "Dilution" method. The leftmost pipette tip is currently down.*

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<sup>10</sup> VENUS 6.0.1, as of April 13, 2024; any VENUS 6 sub-version includes a RESTful API



*Image 39. Credit: Caleb Gibson. An uncropped version of image ff showing the laptop connected to the liquid Handler.*



*Image 40. A closer-up view of how the "Dilution.med" method file looks. The MVP, if done with the liquid handler, is mainly concerned with simply running the method, not defining it.*

While the Hamilton VENUS Programmer's Manual is more for using the VENUS software than it is for interacting with it and using it in conjunction with external sources, the team member who read it was able to gather from it that a potential integration with the VENUS software would require installing any third party software in the "Hamilton\Library\" directory on the liquid handler laptop. Any methods invoking software from a third-party

library need to be linked by saving the library as a "\*.hsl" file and linking it through the "Method → Libraries" menu in the UI of the VENUS software. This could be useful in the event that a library needs to be created, even if only to have it run a connection point for LabVoice to make API calls to. It has also explained how the liquid handler can serve as a temporary application host with the name, "localhost\HAMILTON", for a Microsoft SQL server stored locally on the Windows machine. This database contains data relevant to the methods run by the liquid handler and is called, "HamiltonVectorDB", by default.

Also notably, the manual details error configuration options. Among these is an option to, for any error encountered, send an email to a specified email address. This is less versatile, but presents another design avenue. There is also an option for setting up automatic error handling, which allows the programmer to define a sequence of default methods to run if the instrument encounters an error while running a method (that is, a "\*.hsl" or a "\*.med" file). The manual elucidates how the error-handling can even be configured for one error-handling protocol per method step, and how one can customize which error is flagged in the trace file (for debugging). However, it should be noted that this Programmer's Manual is for version 6 of the VENUS software, not necessarily VENUS 6.0.1. Having an API greatly facilitates direct instrument integration with a LabFlow, so even if the Programmer's Manual is outdated, it will be worthwhile to have VENUS 6.0.1. The Programmer's Manual also elaborates upon method creation with VENUS, should that need to become part of the voice assistant integration's design.

The documentation held valuable information about the VENUS 6 such as several notable software tools tailored to optimize laboratory operations such as components like:

**VENUS Dynamic Scheduler:** This is a software tool for organizing and controlling the workflows of a laboratory equipped with Hamilton and other manufacturers' instruments. By offering control over workflow organization and execution, the VENUS Dynamic Scheduler streamlines operational processes, enhancing efficiency and productivity within the lab.

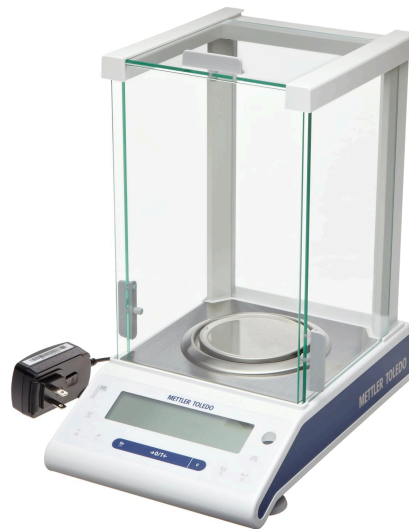
**VENUS Database Plus:** Another important component of the VENUS software suite is VENUS Database Plus. This is a software tool that allows tracking over multiple runs which is also used when data is put in operation from a remote server. This creates a streamlined data management process which ensures accuracy, reliability, and traceability throughout the research process.

VENUS is a Windows-based software solution that has a menu-driven interface allowing the user to define deck layouts and methods as well as executing operations on the Microlab instrument. VENUS gives the user the ability to program and run various different methods for aspirating and dispensing liquids, including control to the accessories such as a wash station and vacuum systems. All programmed methods should be validated by the programmer. The importance of having accurate and reliable experimental outcomes has always been emphasized throughout this project. The VENUS software is most useful when implemented with an instrument like the Hamilton Liquid Handler.

It has thus far been detailed by Mr. Chris Angelelli that the Hamilton Microlab NIMBUS liquid handler can ultimately accept .csv files as instructions for its operations. Although special file extensions are used to create methods for the machine, and then to run or execute those methods, it seemingly may ultimately be capable of responding to merely a .csv file. Although, it is extremely likely that the contents of this .csv file are rather complex and would require a great deal of technical knowledge and acute affiliation with the machine and sophistication in order to be used to run and to execute and to properly instruct the machine as the users desire. Nevertheless, consequently, Mr. Angelelli has thus far suggested that it would be possible to feed the Hamilton Microlab NIMBUS automated liquid handler a .csv file containing a set of instructions, then feed it back to an AI, which has been pre-programmed with, or prompted to, command the liquid handler to do some thing, the results of the input .csv file, that is, and then have the AI evaluate those results. Then, generate a new .csv file to execute something closer to the desired outcome for the instructions given to the liquid handler for the liquid handler to achieve and have it use that new AI-generated .csv file to run and execute and instruct the machine to do something then feed back the results to the AI again in a cycle that does not end until something close enough to the desired results is achieved, as determined by some metric or guideline that is likely related to the particular instruction set or command on hand. However, in practice, Mr. Angelelli has noted (this sort of AI feedback loop cycle for .csv instruction generation was presented as a desire to him by a client), this has proven nearly infeasible if not impossible to achieve, and it is likely that the team will be unable to implement such a mechanism as part of this project due to the scale of work that would be involved.

Nevertheless, despite all of the difficulties of that, it would present a good stretch goal to achieve. It allows the team to, if the MVP and all other more important goals have been accomplished, dabble in further customizing AI as part of an AI integration into a laboratory setting to improve efficiency and eliminate the need for some tedious human labor.

## Mettler-Toledo Digital Balance

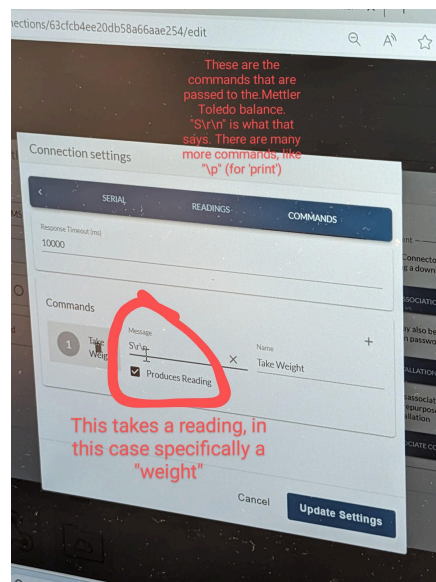


*Image 41. Credit: Mettler-Toledo, Amazon.com*

*(<https://www.amazon.com/Mettler-Toledo-NewClassic-Analytical-Readability/dp/B003P9VQCY>; [https://m.media-amazon.com/images/I/8163vxTwZbL.\\_SX522\\_.jpg](https://m.media-amazon.com/images/I/8163vxTwZbL._SX522_.jpg)). A Mettler-Toledo digital balance similar in functionality, price point, and appearance to the one the sponsor has in their lab.*

According to Mr. Angelelli, one of the simplest instruments to integrate the voice assistant with should be the sponsor's Mettler-Toledo digital balance. This balance is mostly used solely for taking precise measurements. While Mr. Angelelli has indicated there are over 200 possible commands that this balance can be given by a Scitara connection, any one of them would be

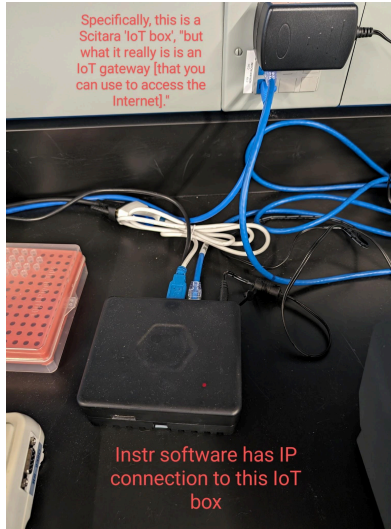
acceptable to Accenture and their clients for demonstrating remote instrument connection and voice-commanded operation.



*Image 42. Credit: Caleb Gibson. An example of using Scitara to command the balance. The circled part is a string for commands to the balance that dictate how its reading appears.*

This balance communicates using the MT-SICS protocol, per Scitara's "METTLER TOLEDO Balance connector" connector description. To communicate with Scitara, this balance relies on a Scitara IoT gateway, since it uses serial communications for Internet connections.





*Image 43. Credit: Caleb Gibson. The IoT gateway box used to enable communication from the Mettler-Toledo balance to Scitara.*

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# **Appendix**

## **Appendix A**

List of Llama 2 LLM fine tuned used for demonstration in image 15

Synthia 70B-v1.2b (<https://huggingface.co/migtissera/Synthia-70B-v1.2b>),  
Euryale-1.3-L2-70B (<https://huggingface.co/Sao10K/Euryale-1.3-L2-70B>),  
Xwin-LM-70B-V0.1 (<https://huggingface.co/Xwin-LM/Xwin-LM-70B-V0.1>). All  
models were put together in interleaving layers using mergekit  
(<https://github.com/arcee-ai/mergekit>) to create a 120 layer model with 103  
billion parameters and quantized to 3.35 bits per weight to fit into 48 GB  
VRAM using ExLlama2.